

1 Differentiable Manifolds

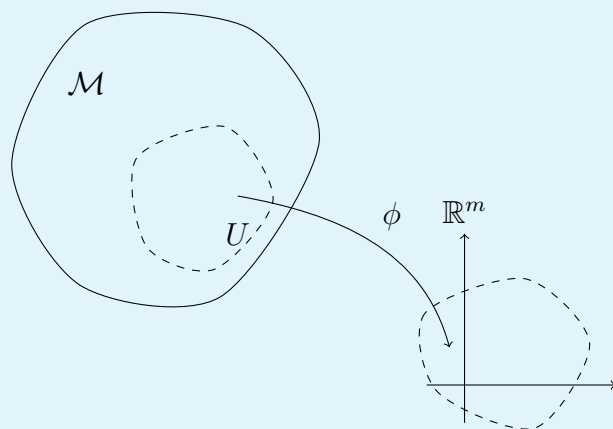
1.1 Definition

1.1.1 Coordinate Charts

Definition 1.1 (Coordinate Charts).

An m -dimensional, $m \neq \infty$ coordinate chart on a topological space $\mathcal{M}$ is a pair

$$(U, \phi) \begin{cases} U \subseteq \mathcal{M}, U \text{ open} \\ \phi : U \rightarrow \mathbb{R}^m, \phi \text{ homeomorphism} \end{cases}$$



✍ Remark.

If $U = \mathcal{M}$, then we say the coordinate chart ϕ is globally defined; if not, then it is locally defined. Few manifolds have globally defined property. $\square$

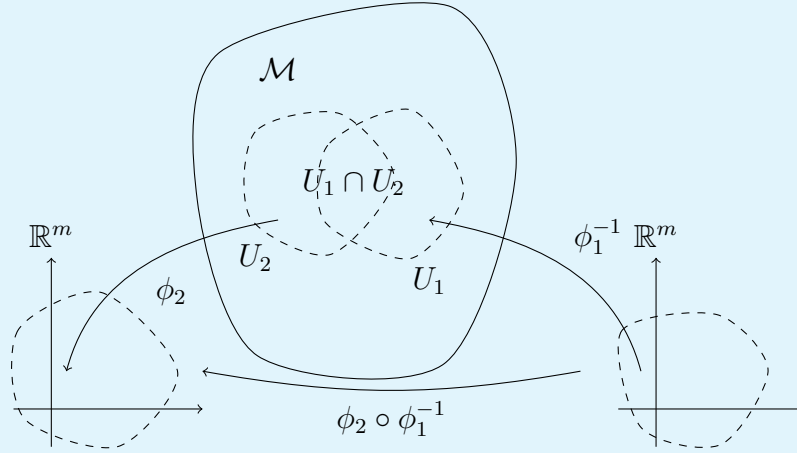
✍ Remark.

The basic method of studying manifolds is to analyze it in the familiar Euclidean space via coordinate charts. $\square$

Definition 1.2 (Overlap Function).

Let $(U_1, \phi_1), (U_2, \phi_2)$ be a pair of m -dimensional coordinate charts with $U_1 \cap U_2 \neq \emptyset$. Then the overlap function is defined as

$$\phi_2 \circ \phi_1^{-1} : \phi_1(U_1 \cap U_2) \subseteq \mathbb{R}^m \rightarrow \phi_2(U_1 \cap U_2) \subseteq \mathbb{R}^m.$$

**Definition 1.3 (Atlas).**

An m -dimensional atlas on $\mathcal{M}$ is a family of m -dimensional coordinate charts $(U_i, \phi_i), i \in I$ s.t.

1. $\mathcal{M} = \bigcup_{i \in I} U_i$.
2. Each overlap function $\phi_j \circ \phi_i^{-1}, i, j \in I$ is C^∞ .

Definition 1.4 (Differentiable Manifolds).

An m -dimensional differentiable manifold is a topological space $\mathcal{M}$ equipped with an atlas.

✂ Remark.

We didn't define a differentiable manifold by regulating the differentiability of the coordinate charts themselves. That's because differentiation is not defined on a manifold, so we need to rely on Euclidean spaces. □

1.2 Dimension of a Manifold

✂ **Remark.**

Consider a manifold that consists of a rod attached to a disk. The dimension is not same everywhere. We give a criterion on how to describe such a scenario.

⌘

Theorem 1.1 (Invariance of Domain).

For all $A, B \subseteq S^n$, if $\exists f : A \rightarrow B$ homeomorphic and $B \in \tau_{S^n}$, then $A \in \tau_{S^n}$ too.

✂ **Remark.**

Theorem 1.1 is an early theorem in algebraic topology.

⌘

Corollary 1.1.1 (Dimension is Well-defined).

Given $U \in \tau_{\mathbb{R}^n}, U' \in \tau_{\mathbb{R}^{n'}}$, and if $\exists f : U \rightarrow U'$ homeomorphic, then $n = n'$.

Proof. If $n = n'$, it is trivially true.

If $n < n'$, embed $\mathbb{R}^n$ to $\mathbb{R}^{n'}$ by $f : \mathbf{x} \mapsto (\mathbf{x}, \mathbf{0})$. Via stereographic projection, we can map homeomorphically

$$\begin{aligned}\phi : U &\mapsto V \subseteq S^{n'}, \\ \phi' : U' &\mapsto V' \subseteq S^{n'}.\end{aligned}$$

Since the compositions are also homeomorphic, we see V and V' are homeomorphic. However, V' is not an open subset of $\mathbb{R}^{n'}$ because of the 0's, which contradicts Invariance of Domain. $\square$

✂ **Remark.**

Since the definition of a differentiable manifold requires every overlap function to be diffeomorphic, if $U \cap U' \neq \emptyset$, their dimensions must be equal via the above corollary. We can bypass this by demanding $U \cap U' = \emptyset$, as in the rod and disk case.

⌘

Corollary 1.1.2.

If $g : V \rightarrow \mathbb{R}^n$ is a continuous injection and $V \in \tau_{\mathbb{R}^n}$, then $g(V)$ is homeomorphic to V , and $g(V) \in \tau_{\mathbb{R}^n}$.

Proof. On $g(V)$, g is surjective and therefore a homeomorphism. Use stereographic projection and the result is obvious. $\square$

1.3 Coordinate Functions

Definition 1.5 (Coordinate Functions).

The coordinate functions are the (Euclidean) components of coordinate.

$$\begin{aligned} \phi : U &\rightarrow \mathbb{R}^m & p &\mapsto \phi(p), \\ \phi^\mu : U &\rightarrow \mathbb{R} & \text{s.t. } \phi(p) &= \begin{pmatrix} \phi^1(p) \\ \vdots \\ \phi^m(p) \end{pmatrix}. \end{aligned}$$

An alternative notation is

$$x^\mu := \phi^\mu.$$

✍ Remark.

There are (Euclidean) projection functions,

$$u^\mu : \mathbb{R}^m \rightarrow \mathbb{R}.$$

But I think mention it will cause a lot of confusion. Just remember in the future when we say $\frac{\partial}{\partial u^\mu}$, we are referring to the Euclidean partial derivative wrt the μ -th component. $\mathfrak{M}$

Definition 1.6 (Jacobian Matrix of Coordinate Transformation).

Let $\mathcal{M}$ be a C^∞ manifold of $\dim \mathcal{M} = m$. Choose two coordinate functions

$$\begin{aligned} \phi : U_\phi &\rightarrow V_\phi, p \mapsto (x^1(p), \dots, x^m(p)), \\ \psi : U_\psi &\rightarrow V_\psi, p \mapsto (y^1(p), \dots, y^m(p)). \end{aligned}$$

Define the Jacobian matrix of coordinate transformation to be

$$J := \begin{pmatrix} \frac{\partial y^1}{\partial x^1} & \cdots & \frac{\partial y^1}{\partial x^m} \\ \vdots & \ddots & \vdots \\ \frac{\partial y^m}{\partial x^1} & \cdots & \frac{\partial y^m}{\partial x^m} \end{pmatrix}$$

In short,

$$J^\nu{}_\mu := \frac{\partial y^\nu}{\partial x^\mu},$$
$$(J^{-1})^\nu{}_\mu := \frac{\partial x^\nu}{\partial y^\mu},$$

where $\frac{\partial y^\nu}{\partial x^\mu} := \frac{\partial(y^\nu \circ \phi^{-1})}{\partial x^\mu}$.

1.4 Manifold With Boundary

1.4.1 Generalized Coordinate Charts

Definition 1.7 (Generalized Coordinate Charts).

A generalized coordinate chart allows chart that

$$\phi : U \rightarrow \phi(U) \subseteq (-\infty, 0] \times \mathbb{R}^{n-1},$$

U is open, and ϕ is homeomorphic.

✍ Remark.

Essentially, this allows a chart to map to "half planes". In this case, even if a set $\phi(U)$ contains $\{0\} \times \mathbb{R}^{n-1}$ and therefore not open in the Euclidean topology of $\mathbb{R}^n$, it is still considered open in the product topology of $(-\infty, 0] \times \mathbb{R}^{n-1}$. $\square$

Definition 1.8 (Manifold With Boundary).

A manifold with boundary is a manifold whose atlas consists of generalized coordinate charts.

Definition 1.9 (Boundary Points of a Manifold).

For all $p \in \mathcal{M}$ is a boundary point of a manifold with boundary $\mathcal{M}$ if $\exists \phi_\alpha \in \Phi$ atlas s.t. $\phi_\alpha^1(p) = 0$.

The set of all boundary points of $\mathcal{M}$ is denoted $\partial\mathcal{M}$.

1.4.2 Boundary is Well-defined

✍ Remark.

A natural question regarding **Definition 1.9** is that, the definition only asks for existence, but it does not guarantee that two coordinate charts will always agree on whether their first component is zero. We resolve this in the following.

$\square$

Theorem 1.2.

Suppose U, U' are open sets in the product topology $(-\infty, 0] \times \mathbb{R}^{n-1}$, and $\exists f : U \rightarrow U'$ homeomorphic. Then

$$f(U \cap (\{0\} \times \mathbb{R}^{n-1})) = U' \cap (\{0\} \times \mathbb{R}^{n-1}).$$

Proof. We show instead that

$$f(U \cap ((-\infty, 0) \times \mathbb{R}^{n-1})) = U' \cap ((-\infty, 0) \times \mathbb{R}^{n-1}).$$

Via **Corollary 1.1.2**, we see $f(U \cap ((-\infty, 0) \times \mathbb{R}^{n-1}))$ must be an open set in the Euclidean topology of $\mathbb{R}^n$. Therefore, it cannot contain $\{0\} \times \mathbb{R}^{n-1}$. $\square$

Corollary 1.2.1 (Boundary is Well-defined).

$\forall p \in \mathcal{M}$, if $\exists \phi_\alpha^1(p) = 0$, then $\forall \phi \in \Phi$ atlas, $\phi^1(p) = 0$.

Proof. Make use of the fact that $\phi \circ \phi_\alpha^{-1}$ is a homeomorphism. $\square$

1.5 Submanifolds

1.5.1 Definition

Definition 1.10 (Local Submanifold).

$A \subseteq \mathcal{M}$ is a submanifold of codimension r around $p \in A \subseteq \mathcal{M}$ if there exists a chart $\phi : U \rightarrow \phi(U) \in \Phi$ atlas s.t. $p \in U$ and

$$\phi(U \cap A) = \phi(U) \cap (\mathbb{R}^{m-r} \times \underbrace{\{0, \dots, 0\}}_{r \text{ zeroes}}).$$

Definition 1.11 (Submanifold).

If A is a C^k local submanifold of $\mathcal{M}$ of codimension r around every $p \in \mathcal{M}$, then we say A is a submanifold of $\mathcal{M}$ of codimension r .

1.5.2 Dimension of Submanifolds

Definition 1.12 (Regular Value).

Let $\mathcal{M}, \mathcal{N}$ be C^∞ manifolds, and $f : \mathcal{M} \rightarrow \mathcal{N}$ be C^∞ . We say $q \in \mathcal{N}$ is a regular value of f if $\forall p \in \tilde{f}(q)$,

$$\begin{cases} \text{rank } f_* : T_p \mathcal{M} \rightarrow T_{f(p)} \mathcal{N} = \dim \mathcal{N} \\ \text{rank } f_* : T_p \partial \mathcal{M} \rightarrow T_{f(p)} \mathcal{N} = \dim \mathcal{N} \quad \text{if } p \in \partial \mathcal{M}. \end{cases}$$

Theorem 1.3 (Regular Value and Submanifolds - With Boundary).

Let $f : \mathcal{M} \rightarrow \mathcal{N}$ be C^∞ . If $q \in \mathcal{N}$ is a regular value of f , then $\tilde{f}(q) = \emptyset$ or a C^∞ submanifold of $\mathcal{M}$ of codimension $\dim \mathcal{N}$. Also, $\partial(\tilde{f}(q)) = \tilde{f}(q) \cap \partial \mathcal{M}$.

Theorem 1.4 (Equation and Submanifolds).

Let $f : \mathcal{M} \rightarrow \mathcal{N}$ be C^∞ . If $\partial \mathcal{M} = \emptyset$ and $\text{rank } f_* = r$ for all $p \in \mathcal{M}$, then for all $q \in \mathcal{N}$, $\tilde{f}(q) = \emptyset$ or a C^∞ submanifold of $\mathcal{M}$ of codimension r .

1.5.3 Dimension of Submanifolds of General Linear Group**Theorem 1.5 (Dimension of Special Linear Group $SL(n, \mathbb{R})$).**

$SL(n, \mathbb{R})$ is a submanifold of $GL(n, \mathbb{R})$ codimension 1; $\dim SL(n, \mathbb{R}) = n^2 - 1$.

Proof. Consider $f = \det : \mathcal{M} = GL(n, \mathbb{R}) \rightarrow \mathbb{R}$. We want to show 1 is a regular value of $f = \det$.

(Elementary argument) One can simply write f_* in local coordinates and work out its rank. First choose a curve $A(t) : (-\epsilon, \epsilon) \rightarrow \mathcal{M}$ s.t. $A(0) = A_0$. Also define $\sigma(t) = \det A(t) : (-\epsilon, \epsilon) \rightarrow \mathbb{R}$. Then $\det_*([A]) \in T_{\det A_0} \mathbb{R} \simeq \mathbb{R}$. Referring to [Theorem 2.11](#), and identify $(\partial_1)_{\det A_0} \in T_{\det A_0} \mathbb{R}$ with 1, we see

$$\det_*([A]) = \left. \frac{\partial \det A}{\partial x^{ij}} \right|_{A=A_0} A'_{ij}(0) = (\text{adj}(A_0))_{ji} A'_{ij}(0).$$

Clearly it didn't vanish everywhere, so it has rank 1. So 1 is indeed a regular value of f , and so by [Theorem 1.3](#), the statement is true.

(Lie argument) Consider the following diagram, which leverages the Lie group property of $SL(n, \mathbb{R})$.

So we know

$$f = \ell_{\det A_0}^{-1} \circ f \circ \ell_{A_0}.$$

We know that both $\ell_{A_0} : A \mapsto A_0 A$ and $\ell_{\det A_0} : a \mapsto a \det A_0$ are linear and thus C^∞ diffeomorphisms. Therefore,

$$\begin{aligned} \text{rank } f_{*A_0} &= \text{rank}(\ell_{\det A_0}^{-1} \circ f \circ \ell_{A_0}^{-1})_{*A_0} \\ &= \text{rank}(\ell_{\det A_0}^{-1} \circ f)_{*I} \\ &= \text{rank}(f_{*I}). \end{aligned}$$

Then one can use the elementary argument again, but this time with significantly less complexity. □

Theorem 1.6 (Dimension of Orthogonal Group $O(n, \mathbb{R})$).

The orthogonal group $O(n, \mathbb{R}) = \{A \in M_n(\mathbb{R}) \mid AA^\top = I_n\}$ is a C^∞ submanifold of $GL(n, \mathbb{R})$ of codimension $\frac{n(n+1)}{2}$, and thus $\dim O(n, \mathbb{R}) = \frac{n(n-1)}{2}$.

2 Tangent Spaces

✍ Remark.

The definition of manifold do not require the entity to be embeded in a higher dimensional space. Therefore, the traditional view of tangency is not valid here.

✍

2.1 Curves and Vectors

Definition 2.1 (Curve).

A curve on $\mathcal{M}$ is a C^∞ map,

$$\sigma : (-\epsilon, \epsilon) \rightarrow \mathcal{M}.$$

Definition 2.2 (Curve Tangency).

Two curves σ_1, σ_2 are tangent at $p \in \mathcal{M}$ if

1. $\sigma_1(0) = \sigma_2(0) = p$.
2. $\frac{d}{dt}(x^i \circ \sigma_1(t)) \Big|_{t=0} = \frac{d}{dt}(x^i \circ \sigma_2(t)) \Big|_{t=0}, \quad 1 \leq i \leq m.$

✍ Remark.

Written more compactly,

$$\frac{d}{dt}(\phi \circ \sigma_1) \Big|_{t=0} = \frac{d}{dt}(\phi \circ \sigma_2) \Big|_{t=0}$$

✍

Definition 2.3 (Tangent Vectors).

A tangent vector at $p \in \mathcal{M}$ is an equivalence class of curves where the equivalence relation is that they are tangent. It will be denoted as

$$v = [\sigma]_{t=t_0}, \quad v \in T_{\sigma(t_0)}\mathcal{M}.$$

If the time is not specified, it will be understood as $t = 0$.

Definition 2.4 (Tangent Space).

The tangent space $T_p\mathcal{M}$ at point p is the set of all tangent vectors at point p .

2.2 Curves and Derivation

Definition 2.5 (Directional Derivative).

For any $f : \mathcal{M} \rightarrow \mathbb{R}$ s.t. $f \in C^\infty$, we define

$$v(f) := \left. \frac{d}{dt}(f \circ \sigma(t)) \right|_{t=0},$$

where $v = [\sigma]$.

Theorem 2.1.

The definition **Definition 2.5** is well-defined. That is, $v(f)$ is independent of the curve σ chosen as well as $v = [\sigma]$.

Proof. Let $v_1 = [\sigma_1] = [\sigma_2] = v_2$. Then

$$\begin{aligned} v_1(f) &= \left. \frac{d}{dt}(f \circ \sigma_1) \right|_{t=0}, \\ v_2(f) &= \left. \frac{d}{dt}(f \circ \sigma_2) \right|_{t=0}, \\ \frac{d}{dt}(f \circ \sigma_1) \Big|_{t=0} &= \frac{d}{dt}(f \circ \sigma_2) \Big|_{t=0}. \end{aligned}$$

Then

$$\begin{aligned}
v_1(f) &= \left. \frac{d}{dt} \left(\underbrace{(f \circ \phi^{-1})}_{\mathbb{R} \leftarrow \mathbb{R}^m} \circ \underbrace{(\phi \circ \sigma_1)}_{\mathbb{R}^m \leftarrow \mathbb{R}} \right) \right|_{t=0} \\
&= (f \circ \phi^{-1})' \circ (\phi \circ \sigma_1) \cdot (\phi \circ \sigma_1)' \Big|_{t=0} \\
&= (f \circ \phi^{-1})' \circ (\phi \circ \sigma_2) \cdot (\phi \circ \sigma_2)' \Big|_{t=0} \\
&= v_2(f),
\end{aligned}$$

since $\phi \circ \sigma_1(0) = \phi \circ \sigma_2(0) = \phi(p)$, and $(\phi \circ \sigma_1)' = (\phi \circ \sigma_2)'$ by equivalence. $\square$

2.3 Isomorphism with Euclidean Spaces

Definition 2.6 ("Straight Lines").

Choose a coordinate chart $\phi \in \Phi$ atlas near p . For all $v \in \mathbb{R}^m$, we define

$$\gamma_v^\phi(t) := \phi^{-1}(\phi(p) + tv).$$

In simple words, it is such a curve on manifold that it is a straight line on maps.

Theorem 2.2 (Isomorphism with Straight Lines).

Let $p \in \mathcal{M}$, $p \in \phi \in \Phi$. For any curve γ passing through p , $\exists! v \in \mathbb{R}^m$ s.t. γ is tangent to γ_v^ϕ , where v can be explicitly given by $(\phi \circ \gamma)'(0)$.

In other words, the map

$$\begin{aligned}
\ell_p^\phi : \mathbb{R}^m &\rightarrow T_p \mathcal{M} \\
v &\mapsto [\gamma_v^\phi]
\end{aligned}$$

is a bijection with inverse

$$\begin{aligned}
(\ell_p^\phi)^{-1} : T_p \mathcal{M} &\rightarrow \mathbb{R}^m \\
[\gamma] &\mapsto (\phi \circ \gamma)'(0).
\end{aligned}$$

Proof. It is almost by definition that γ is tangent to $(\phi \circ \gamma)'(0)$.

Suppose v_1, v_2 satisfies γ is tangent to $\gamma_{v_1}^\phi, \gamma_{v_2}^\phi$. Then they are tangent too. So

$$(\phi \circ \gamma_{v_1}^\phi)'(0) = (\phi \circ \gamma_{v_2}^\phi)'(0).$$

Therefore, $v_1 = v_2$. $\square$

Definition 2.7 (Alternative Definition for Addition).

For $v_1, v_2 \in T_p\mathcal{M}$, we define addition via the isomorphism **Theorem 2.2**,

$$v_1 + v_2 := (\ell_p^\phi)^{-1}(\ell_p^\phi(v_1) + \ell_p^\phi(v_2)).$$

Definition 2.8 (Alternative Definition of Basis Tangent Vectors).

$$(\partial_\mu)_p := \ell_p^\phi(e^\mu).$$

Theorem 2.3.

$$(\partial_\mu)_p(x^\nu) = \delta_\mu^\nu.$$

Proof. By **Definition 2.8**,

$$(\partial_\mu)_p := \ell_p^\phi(e^\mu) = \phi^{-1}(\phi(p) + te^\mu),$$

where e^μ is the column vector with μ -th component 1, others 0. Then

$$\begin{aligned} (\partial_\mu)_p(x^\nu) &= (x^\nu \circ \phi^{-1}(\phi(p) + te^\mu))'(0) \\ &= (x^\nu \circ \phi^{-1})'(\phi(p))e^\mu. \end{aligned}$$

But since x^ν has only one component, $(x^\nu \circ \phi^{-1})'$ is a row vector with ν -th component 1, others 0. So the result follows immediately. $\square$

Theorem 2.4 (Linear Independence of Basis Tangent Vectors).

The basis tangent vectors $(\partial_\mu)_p, 1 \leq \mu \leq \dim \mathcal{M}$ are linear independent.

Proof. Suppose $a^\mu (\partial_\mu)_p = 0$. Then

$$a^\mu (\partial_\mu)_p(x^\nu) = a^\mu \delta_\mu^\nu = 0(x^\nu) = 0.$$

So $a^\nu = 0$. $\square$

Theorem 2.5 (Coordinate Expansion of Tangent Vectors).

For all $v \in T_p\mathcal{M}$, we have

$$v = v^\mu (\partial_\mu)_p,$$

where Einstein notation was used, and $v^\mu = v(x^\mu)$.

✍ Remark.

A natural question is that whether two charts behave "the same" if $U_1 \cap U_2 \neq \emptyset$. Under this perspective, the criterion is clear: we need only to check whether a straight line is still a straight line in another chart, which is true indeed. It also produces the coordinate transformation formula for free. See below. $\square$

Theorem 2.6 (Coordinate Transformation of Straight Lines).

Choose $\phi, \psi \in \Phi$, $\phi : U_1 \rightarrow \mathbb{R}^m$, $\psi : U_2 \rightarrow \mathbb{R}^m$ s.t. $U_1 \cap U_2 \neq \emptyset$ and $p \in U_1 \cap U_2$. Let the corresponding straight line isomorphisms ℓ_p^ϕ, ℓ_p^ψ . Then $(\ell_p^\psi)^{-1} \circ \ell_p^\phi$ is a linear isomorphism.

Let the local coordinates induced by ϕ be $x^1, \dots, x^m$, ψ be $y^1, \dots, y^m$, then $(\ell_p^\psi)^{-1} \circ \ell_p^\phi$ can be expressed in terms of the Jacobian **Definition 1.6** $J^\nu_\mu := \frac{\partial y^\nu}{\partial x^\mu} \Big|_{\phi(p)}$, namely,

$$(\ell_p^\psi)^{-1} \circ \ell_p^\phi(v) = Jv.$$

Proof.

$$\begin{aligned} (\ell_p^\psi)^{-1} \circ \ell_p^\phi(v) &= (\ell_p^\psi)^{-1}(\phi^{-1}(\phi(p) + vt)) \\ &= ((\psi \circ \phi^{-1})(\phi(p) + vt))'(0) \\ &= Jv \end{aligned}$$

$\square$

Definition 2.9 (Contravariancy and Covariancy).

Let $\mathcal{M}$ be a m -dimensional C^∞ manifold. Choose two coordinate charts

$$\begin{aligned} \phi : U_\phi &\rightarrow V_\phi, p \mapsto (x^1(p), \dots, x^m(p)), \\ \psi : U_\psi &\rightarrow V_\psi, p \mapsto (y^1(p), \dots, y^m(p)). \end{aligned}$$

and the corresponding Jacobian matrix **Definition 1.6** $J^\nu_\mu := \frac{\partial y^\nu}{\partial x^\mu} \Big|_{\phi(p)}$. We

define covariancy to be anything that transforms like

$$(\text{new})_\nu = (\text{old})_\mu (J^{-1})^\mu{}_\nu.$$

Corollary 2.6.1 (Contravariancy of Tangent Vectors).

The basis vectors transform covariantly, whereas the components of vectors transform contravariantly. That is, choose two overlapping coordinate charts $(x^1, \dots, x^m)$ and $(y^1, \dots, y^m)$ and define their Jacobian matrix **Definition 1.6**

$$J^\nu{}_\mu := \left. \frac{\partial y^\nu}{\partial x^\mu} \right|_{\phi(p)},$$

$$\begin{cases} \left(\frac{\partial}{\partial y^\nu} \right) = \left(\frac{\partial}{\partial x^\mu} \right) (J^{-1})^\mu{}_\nu & (\text{covariant}) \\ v^{\nu'} = J^{\nu'}{}_\mu v^\mu & (\text{contravariant}) \end{cases}$$

2.4 Pushforward

2.4.1 Definition and Linearity

✍ Remark.

The pushforward $h_* : T_p \mathcal{M} \rightarrow T_{h(p)} \mathcal{N}$ of a specific function $h : \mathcal{M} \rightarrow \mathcal{N}$ can be thought of as local linearization of the function.

□

Definition 2.10 (Pushforward).

Given a function $h : \mathcal{M} \rightarrow \mathcal{N}$ and $v \in T_p \mathcal{M}$, then we define the pushforward $h_* : T_p \mathcal{M} \rightarrow T_{h(p)} \mathcal{N}$ by

$$h_*(v) := [h \circ \sigma], \quad v = [\sigma].$$

Theorem 2.7.

The pushforward operation **Definition 2.10** is well-defined. That is, $h_*(v_1) = h_*(v_2)$ if $v_1 = [\sigma_1] = [\sigma_2] = v_2$.

Theorem 2.8 (Algebraic Definition of Pushforward).

The definition of pushforward **Definition 2.10** is equivalent to the following: let

$h : \mathcal{M} \rightarrow \mathcal{N}$, $h_* : D_p \mathcal{M} \rightarrow D_{h(p)} \mathcal{M}$ is defined by,

$$(h_* v)(f) := v(f \circ h).$$

Proof. ($\rightarrow$)

$$\begin{aligned} h_*(v)(f) &= [h \circ \sigma](f) = \frac{d}{dt}(f \circ h \circ \sigma(t)) \Big|_{t=0} \\ &= \frac{d}{dt}((f \circ h) \circ \sigma(t)) \Big|_{t=0} \\ &:= v(f \circ h). \end{aligned}$$

($\leftarrow$) This direction is similar. □

Theorem 2.9 (Linearity of Pushforward).

The pushforward map $h_* : T_p \mathcal{M} \rightarrow T_{h(p)} \mathcal{N}$ is linear.

$$h_*(rv_1 + v_2) = rh_*(v_1) + h_*(v_2).$$

Proof. (Using **Definition 2.10**) Let $p \in (U, \phi) \subseteq \mathcal{M}$, and $h(p) \in (V, \psi) \subseteq \mathcal{N}$. Choose ϕ s.t. $\phi(p) = 0$. It is obvious that $h_*(rv_1 + v_2)(0) = (rh_*(v_1) + h_*(v_2))(0) = h(p)$.

Consider

$$\begin{aligned} \frac{d}{dt} \underbrace{(\psi \circ h_*(rv_1 + v_2))}_{\mathbb{R}^n \leftarrow \mathcal{N} \leftarrow \mathbb{R}} \Big|_{t=0} &= \frac{d}{dt} \left(\underbrace{\psi \circ h \circ (\phi^{-1})}_{\mathbb{R}^n \leftarrow \mathcal{N} \leftarrow \mathcal{M} \leftarrow \mathbb{R}^m} \circ \underbrace{(r\phi \circ \sigma_1 + \phi \circ \sigma_2)}_{\mathbb{R}^m \leftarrow \mathcal{M} \leftarrow \mathbb{R}} \right) \Big|_{t=0} \\ &= (\psi \circ h \circ \phi^{-1})' \circ (r\phi \circ \sigma_1 + \phi \circ \sigma_2) \cdot (r\phi \circ \sigma_1 + \phi \circ \sigma_2)' \Big|_{t=0} \\ &= (\psi \circ h \circ \phi^{-1})'(0) \cdot ((r\phi \circ \sigma_1)' + (\phi \circ \sigma_2)') \Big|_{t=0}. \end{aligned}$$

And

$$\begin{aligned} \frac{d}{dt} \left(\underbrace{\psi}_{\mathbb{R}^n \leftarrow} \circ \underbrace{(rh_*(v_1) + h_*(v_2))}_{\mathcal{N} \leftarrow \mathbb{R}} \right) \Big|_{t=0} &= \frac{d}{dt} \underbrace{(r\psi \circ h \circ \sigma_1 + \psi \circ h \circ \sigma_2)}_{\mathbb{R}^n \leftarrow \mathcal{N} \leftarrow \mathcal{M} \leftarrow \mathbb{R}} \Big|_{t=0} \\ &= \left(\underbrace{r\psi \circ h \circ \phi^{-1}}_{\mathbb{R}^n \leftarrow \mathcal{N} \leftarrow \mathcal{M} \leftarrow \mathbb{R}^m} \circ \underbrace{\phi \circ \sigma_1}_{\mathbb{R}^m \leftarrow \mathcal{M} \leftarrow \mathbb{R}} + \psi \circ h \circ \phi^{-1} \circ \phi \circ \sigma_2 \right)' \Big|_{t=0} \\ &= (r(\psi \circ h \circ \phi^{-1})' \circ (\phi \circ \sigma_1) \cdot (\phi \circ \sigma_1)') \Big|_{t=0} \\ &\quad + ((\psi \circ h \circ \phi^{-1})' \circ (\phi \circ \sigma_2) \cdot (\phi \circ \sigma_2)') \Big|_{t=0} \\ &= (\psi \circ h \circ \phi^{-1})'(0) \cdot (r(\phi \circ \sigma_1)' + (\phi \circ \sigma_2)') \Big|_{t=0}. \end{aligned}$$

So we see the two are equal.

(Using **Theorem 2.8**)

$$\begin{aligned}
(h_*(rv_1 + v_2))(f) &= (rv_1 + v_2)(f \circ h) \\
&= rv_1(f \circ h) + v_2(f \circ h) \\
&= r(h_*v_1)f + (h_*v_2)f.
\end{aligned}$$

(Using straight line isomorphism: **Definition 2.7**)

$$\begin{array}{ccc}
T_p\mathcal{M} & \xrightarrow{h_*} & T_{h(p)}\mathcal{N} \\
\ell_p^\phi \uparrow & & \downarrow (\ell_{h(p)}^\psi)^{-1} \\
\mathbb{R}^m & \xrightarrow{\quad} & \mathbb{R}^n
\end{array}$$

Since ℓ_p^ϕ and $(\ell_{h(p)}^\psi)^{-1}$ are linear, to prove h_* is linear, we need only to show $(\ell_{h(p)}^\psi)^{-1} \circ h_* \circ \ell_p^\phi$ is linear.

$$\begin{aligned}
&(\ell_{h(p)}^\psi)^{-1} \circ h_* \circ \ell_p^\phi(v) \\
&= (\ell_{h(p)}^\psi)^{-1} \circ h_*([\phi^{-1}(\phi(p) + tv)]) \\
&= (\ell_{h(p)}^\psi)^{-1}([h \circ \phi^{-1}(\phi(p) + tv)]) \\
&= (\psi \circ h \circ \phi^{-1}(\phi(p) + tv))'(0) \\
&= (\psi \circ h \circ \phi^{-1})'(\phi(p))v.
\end{aligned}$$

□

Theorem 2.10 (Associativity of Pushforwards).

Given manifolds $\mathcal{M}, \mathcal{N}, \mathcal{P}$ and $h : \mathcal{M} \rightarrow \mathcal{N}$, $k : \mathcal{N} \rightarrow \mathcal{P}$, then

$$(k \circ h)_* = k_* \circ h_*.$$

2.4.2 Jacobian

Theorem 2.11 (Local Representative of Pushforward).

Let $\dim \mathcal{M} = m$, $\dim \mathcal{N} = n$, $h : \mathcal{M} \rightarrow \mathcal{N}$, $\{x^1, \dots, x^m\}$ be the local coordinates of $\mathcal{M}$ around p , and $\{y^1, \dots, y^n\}$ be the local coordinates of $\mathcal{N}$ around $h(p)$.

Then

$$h_*v = \sum_{\mu=1}^m \sum_{\nu=1}^n (\partial_\nu)_{h(p)} \left. \frac{\partial h^\nu}{\partial x^\mu} \right|_p v^\mu,$$

where $J^\nu_\mu := \left. \frac{\partial h^\nu}{\partial x^\mu} \right|_p := (\partial_\mu)_p (y^\nu \circ h)$ is the Jacobian matrix.

Proof. First expand v in terms of local coordinates and use linearity,

$$h_*v = h_*(v^\mu (\partial_\mu)_p) = v^\mu h_*((\partial_\mu)_p).$$

Expand the result in local coordinates of $\mathcal{N}$,

$$h_*((\partial_\mu)_p) = \left(h_* (\partial_\mu)_p \right)^\nu (\partial_\nu)_{h(p)}.$$

Using **Theorem 2.8**,

$$\begin{aligned} \left(h_* (\partial_\mu)_p \right)^\nu &= \left(h_* (\partial_\mu)_p \right) \circ y^\nu \\ &= (\partial_\mu)_p (y^\nu \circ h) \\ &:= (\partial_\mu)_p h^\nu. \end{aligned}$$

So,

$$h_*((\partial_\mu)_p) = (\partial_\mu)_p h^\nu (\partial_\nu)_{h(p)}.$$

And,

$$h_*v = v^\mu (\partial_\mu)_p h^\nu (\partial_\nu)_{h(p)}.$$

□

Theorem 2.12 (Using Curve to Pushforward).

Given $c : (-\epsilon, \epsilon) \rightarrow \mathcal{M}$ a curve, and choose the coordinate chart of $\mathbb{R}$ to be the identity, then

$$c_* \left(\frac{d}{dt} \right)_0 = [c] \in T_p \mathcal{M}.$$

Proof. First we clarify what is $\left(\frac{d}{dt} \right)_0$. Since on the trivial manifold $\mathbb{R}$ there is only one coordinate, namely t , we need not specify the number. Also, considering our functions are scalar valued $f : \mathcal{M} \rightarrow \mathbb{R}$, this motivates us to write "total differential".

For all $f \in C^\infty$,

$$c_* \left(\frac{d}{dt} \right)_0 f = \left(\frac{d}{dt} \right)_0 (f \circ c).$$

Since the coordinate chart is the identity,

$$\begin{aligned}\left(\frac{d}{dt}\right)_0 (f \circ c) &= \frac{d}{dt}(f \circ c \circ I) \Big|_{I(t)=0} \\ &= \frac{d}{dt}(f \circ c) \Big|_{t=0} \\ &= [c]f.\end{aligned}$$

□

3 Vector Fields and the Tangent Bundle

3.1 Vector Fields

Definition 3.1 (Vector Fields).

A vector field X on $V \subseteq \mathcal{M}$ is any rule of assigning a tangent vector $X(p) = X_p \in T_p\mathcal{M}$ for all $p \in V$.

Definition 3.2 (Components of a Vector Field).

Let X be a vector field on $V \subseteq \mathcal{M}$, $\dim \mathcal{M} = m$. For any chart $\phi : U \rightarrow \phi(U) \in \Phi$ with induced coordinates $x^1, \dots, x^m$ and any $p \in V \cap U$, the decomposition $X(p) = X^j(p) (\partial_j)_p$ is unique, and therefore we write

$$X = X^j \partial_j$$

on $V \cap U$, and $X^j : V \cap U \rightarrow \mathbb{R}$ are called the components of X on $V \cap U$.

Definition 3.3 (Smoothness of a Vector Field).

A vector field X on $V \subseteq \mathcal{M}$ is C^k near $p \in V$ iff $\exists \phi \in \Phi$ s.t. $p \in U_\phi$ and all the components X^j induced by ϕ are C^k . That is,

$$X^j \circ \phi^{-1} \text{ are } C^k \text{ on } \phi(V \cap U).$$

3.2 Tangent Bundle

3.2.1 Definition

Definition 3.4 (Tangent Bundle).

The tangent bundle $T\mathcal{M}$ is

$$T\mathcal{M} := \bigcup_{p \in \mathcal{M}} T_p\mathcal{M}.$$

✍ Remark.

Why not $T\mathcal{M} := \{T_p\mathcal{M} \mid p \in \mathcal{M}\}$?

☞

3.2.2 Projection

Definition 3.5 (Canonical Projection).

The canonical projection is the map

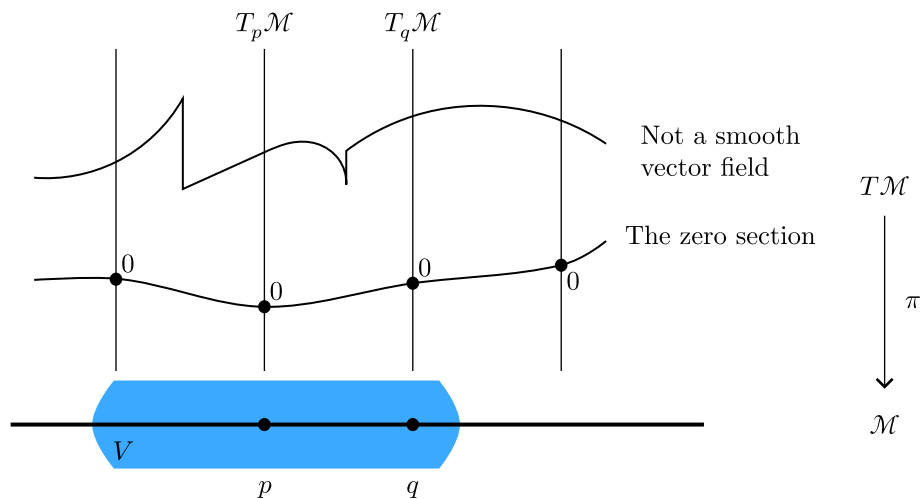
$$\begin{aligned}\pi : T\mathcal{M} &\rightarrow \mathcal{M} \\ T_p\mathcal{M} &\mapsto p.\end{aligned}$$

Definition 3.6 (Alternative Definition of Vector Fields).

A tangent vector field X on $V \subseteq \mathcal{M}$ is a map $X : V \rightarrow T\mathcal{M}$ s.t. $(\pi \circ X)(p) = p$ for all $p \in V$.

Remark.

Let's see why vector fields are often called a "cross-section" of a tangent bundle.



□

3.2.3 Topological and Manifold Structure

Definition 3.7 (Smooth Structure on Tangent Bundle).

Let a manifold $\mathcal{M}$ with dimension m and atlas Φ . Consider a chart $\phi_j \in \Phi : U_j \rightarrow \mathbb{R}^m$. We define a chart $\tilde{\phi}_j$ for the tangent bundle $T\mathcal{M}$ accordingly,

$$\begin{aligned} \tilde{\phi}_j : \tilde{\pi}(U_j) \subseteq T\mathcal{M} &\rightarrow \mathbb{R}^{2m} \\ (p, v^i \partial_i) &\mapsto (\phi_j(p), v^1, \dots, v^m), \end{aligned}$$

where $\tilde{\pi}$ is the inverse set map of the canonical projection function **Definition 3.5**. We thus see the tangent bundle $T\mathcal{M}$ is itself a manifold of dimension $2m$.

Also, we define the topology of the tangent bundle by

$$A \subseteq T\mathcal{M} \text{ is open iff } \tilde{\phi}_j(A \cap \tilde{\pi}(U_j)) \text{ is open.}$$

Theorem 3.1.

The smooth structure given by **Definition 3.7** is unique in the sense of,

1. $\pi : T\mathcal{M} \rightarrow \mathcal{M}$ is C^∞ ,
2. For all open sets $V \subseteq \mathcal{M}$ and any vector field X on V , X is $C^\infty \iff X : V \rightarrow T\mathcal{M}$ is C^∞ .

✂ Remark.

At first thought, one may think of $T\mathcal{M} \simeq \mathcal{M} \times \mathbb{R}^m$. This is not the case, as one can consider the Möbius strip. The tangent bundle is only locally isomorphic to $U \times \mathbb{R}^m$.

If indeed $T\mathcal{M} \simeq \mathcal{M} \times \mathbb{R}^m$, then the bundle is called trivial tangent bundle, and the manifold is called parallelizable. □

3.3 Integral Curves and Local Flows

3.3.1 Integral Curves

Definition 3.8 (Integral Curves).

Let $\mathcal{M}$ be a manifold, and X be a vector field on $V \subseteq \mathcal{M}$. If one single curve $\sigma : (-\epsilon, \epsilon) \rightarrow V$, $\epsilon > 0$ satisfies

$$\begin{aligned}\sigma(0) &= p \in V \\ X_{\sigma(t)} &= [\sigma] \quad \forall t \in (-\epsilon, \epsilon)\end{aligned}$$

then σ is called an integral curve of the vector field X through the point p .

Theorem 3.2 (Differential Equations of Integral Curve).

The components X^μ of X determine the integral curve σ by the following ODE with boundary conditions,

$$\begin{aligned}X^\mu(\sigma(t)) &= \frac{d}{dt}x^\mu(\sigma(t)) \\ x^\mu(\sigma(0)) &= x^\mu(p), \mu = 1, 2, \dots, m.\end{aligned}$$

3.3.2 Local Flows

Definition 3.9 (Local 1D Family of Local Diffeomorphisms).

A local, 1D family of local diffeomorphisms at $p \in \mathcal{M}$ is made up of a family of diffeomorphisms $\{\sigma_t : U \rightarrow \mathcal{M} \mid t \in (-\epsilon, \epsilon)\}$ with $\epsilon > 0$, $U \subseteq \mathcal{M}$ an open set s.t.

1. Every σ_t is a smooth function in t and p .
2. $\forall t, s \in \mathbb{R}$ and $|t|, |s|, |t+s| < \epsilon$, and $\forall p \in U$ s.t. $\sigma_t(p), \sigma_s(p), \sigma_{t+s}(p) \in U$, we have

$$\sigma_s(\sigma_t(p)) = \sigma_{s+t}(p).$$

3. $\sigma_0(p) = p$.

✍ **Remark.**

The first "local" refers to the parameter t , which is limited to $(-\epsilon, \epsilon)$. The second "local" refers to the spatial limitation to U . You can view $\phi_t(q)$ as a curve that brings $t \in (-\epsilon, \epsilon)$ to $\phi_t(q) \in \mathcal{M}$. □

Definition 3.10.

Consider a family of local diffeomorphisms given by **Definition 3.9** ϕ_t . We denote the following for convenience, if its meaning is clear from context,

$$\begin{aligned}\sigma(p, t) &:= \sigma_t(p), \\ \sigma_p(t) &:= \sigma_t(p).\end{aligned}$$

Definition 3.11 (Local Flow).

Let X be a vector field on $V \subseteq \mathcal{M}$, $p \in V$. Consider a family of diffeomorphisms **Definition 3.9** σ on V . Let the corresponding curve family $\sigma_p(t)$ satisfy,

$$\begin{aligned}\sigma_p(0) &= p & \forall p \in V, \\ [\sigma_p] &= X_{\sigma_p(t)} = \sigma_{p*} \left(\frac{d}{dt} \right)_t & \forall t \in (-\epsilon, \epsilon), p \in V.\end{aligned}$$

Then we say the curve family $\sigma_p(t)$ is the local flow for X on V . We may abbreviate and just say σ is the local flow for X .

Theorem 3.3.

Local flows always exist and are unique. Thus for a vector field X defined on V , we may denote its local flow of over an open set U by σ^X .

3.4 Lie Derivative of Vector Fields

3.4.1 Lie Bracket

Definition 3.12 (Lie Derivative of Functions).

We can view a vector field as a function $X : C^\infty \rightarrow C^\infty$, given by

$$(Xf)(p) := X_p f = \left. \frac{\partial}{\partial t} (f \circ \sigma^X(p, t)) \right|_{t=0}.$$

We can interpret this as: How much the function f changes beginning at p , along the flow lines of X ?

Definition 3.13 (Composition of Vector Fields).

We can view $X : C^\infty(\mathcal{M}) \rightarrow C^\infty(\mathcal{M})$, and so does Y . Therefore, we define

$$(X \circ Y)(f) := X(Yf).$$

Theorem 3.4 (Flow Interpretation).

$$X_p(Yf) = \left. \frac{\partial^2}{\partial s \partial t} (f(\sigma^Y(\sigma^X(p, t), s))) \right|_{(s,t)=(0,0)}.$$

We can interpret $X(Yf)$ as: How much the function f changes beginning at the point p , first following the flow lines of X , then following the flow lines of Y ?

Definition 3.14 (Lie Bracket/Vector Field Commutator).

We define the Lie Bracket of two vector fields X, Y to be

$$[X, Y] := X \circ Y - Y \circ X.$$

✍ Remark.

Lie Bracket **Definition 3.14** is a vector field, while the expression $X \circ Y$ is not, because it contains second differential terms. See the following proof. $\square$

Theorem 3.5 (Lie Bracket Components).

$$[X, Y]^\mu = (X^\nu \partial_\nu Y^\mu - Y^\nu \partial_\nu X^\mu).$$

Proof. Given $X = X^\mu \partial_\mu, Y = Y^\nu \partial_\nu$, we try to write the component of $X \circ Y$.

$$X \circ Y(f) = X^\mu \partial_\mu (Y^\nu \partial_\nu f).$$

However, notice that

$$\begin{aligned} Y^\nu &:= Y x^\nu \in C^\infty(\mathcal{M}); \\ \partial_\nu &: C^\infty(\mathcal{M}) \rightarrow C^\infty(\mathcal{M}), \\ &\implies \partial_\nu f \in C^\infty(\mathcal{M}). \end{aligned}$$

So we need to use the Leibniz property of ∂_μ in order to evaluate the second term. Doing this for $X \circ Y(f)$ and $Y \circ X(f)$, we have

$$\begin{aligned} X \circ Y(f) &= X^\mu ((\partial_\mu Y^\nu)(\partial_\nu f) + Y^\nu \partial_\mu \partial_\nu f). \\ Y \circ X(f) &= Y^\nu ((\partial_\nu X^\mu)(\partial_\mu f) + X^\mu \partial_\nu \partial_\mu f). \end{aligned}$$

So if $\partial_\mu \partial_\nu f = \partial_\nu \partial_\mu f$, then by subtracting, we can cancel the second order terms, and we are done. We prove so now.

$$\begin{aligned} (\partial_\mu \partial_\nu f)(p) &= \frac{\partial}{\partial u^\mu} ((\partial_\nu f) \circ \phi^{-1}) \Big|_{\phi(p)} \\ &= \frac{\partial}{\partial u^\mu} ((\partial_\nu)_{\phi^{-1}(u)} f) \Big|_{\phi(p)} \\ &= \frac{\partial}{\partial u^\mu} \left(\frac{\partial}{\partial u^\nu} (f \circ \phi^{-1}) \Big|_u \right) \Big|_{\phi(p)} \\ &= \frac{\partial}{\partial u^\nu} \left(\frac{\partial}{\partial u^\mu} (f \circ \phi^{-1}) \Big|_u \right) \Big|_{\phi(p)} \\ &= (\partial_\nu \partial_\mu f)(p). \end{aligned}$$

□

Theorem 3.6 (Properties of Lie Brackets).

$$\begin{cases} [X, Y] = -[Y, X] & (\text{antisymmetry}) \\ \sum_{\text{cyc}} [X, [Y, Z]] = 0. & (\text{Jacobi Identity}) \end{cases}$$

Theorem 3.7 (Functional Product Rule).

$$[fX, gY] = f(Xg)Y - g(Yf)X + fg[X, Y].$$

Proof. We first deal with the case $[X, fY]$.

$$\begin{aligned} [X, fY] &= X(fY) - (fY)X \\ &= (Xf)Y + fXY - fYX \\ &= (Xf)Y + f[X, Y]. \end{aligned}$$

To expand $X(fY)$ you can expand X into local frame, and use the product (Leibniz) rule on ∂_μ .

Now the general case follows,

$$\begin{aligned} [fX, gY] &= (fXg)Y + g[fX, Y] \\ &= f(Xg)Y - g[Y, fX] \\ &= f(Xg)Y - g((Yf)X + f[Y, X]) \\ &= f(Xg)Y - g(Yf)X + fg[X, Y]. \end{aligned}$$

□

Theorem 3.8 (Pushforward and Lie Bracket).

Given two manifolds $\mathcal{M}, \mathcal{N}$ and a map $h : \mathcal{M} \rightarrow \mathcal{N}$ whose pushforward h_* , if two vector fields on $\mathcal{M}$ have well-defined pushforwards, then

$$h_*[X, Y] = [h_*X, h_*Y].$$

Proof. First of all, we write out explicitly the corresponding points,

$$(h_*Y)_{h(p)} f = Y_p (f \circ h).$$

This means, as functions,

$$Y(f \circ h) = ((h_*Y)f) \circ h.$$

Apply $h_*[X, Y]$ to a function f ,

$$\begin{aligned}
(h_*[X, Y])f &= X(Y(f \circ h)) - Y(X(f \circ h)) \\
&= X\left((h_*Y)f \circ h\right) - Y\left((h_*X)f \circ h\right) \\
&= \left(h_*X((h_*Y)f) - h_*Y((h_*X)f)\right) \circ h \\
&= ([h_*X, h_*Y]f) \circ h
\end{aligned}$$

which is actually the desired result, for

$$h_*([X, Y]_p) = [h_*X, h_*Y]_{h(p)}.$$

□

3.4.2 Lie Derivative of Vector Fields

Definition 3.15 (Lie Derivative).

Let X, Y be two vector fields defined on $U, V \subseteq \mathcal{M}$, respectively. Then the lie derivative of Y along X is defined as

$$(\mathcal{L}_X Y)_p := \lim_{t \rightarrow 0} \frac{\sigma_{-t}^X(Y_{\sigma_t^X(p)}) - Y_p}{t},$$

where the latter is often abbreviated as

$$\left. \frac{d}{dt}(\sigma_{-t}^X(Y_{\sigma_t^X(p)})) \right|_{t=0} = \lim_{t \rightarrow 0} \frac{\sigma_{-t}^X(Y_{\sigma_t^X(p)}) - Y_p}{t}.$$

✍ **Remark.**

It is common to interpret the Lie derivative of Y along X to be: How much do Y change when one moves along the flow lines of X ?

However, one cannot directly compare the two vectors $Y_{\sigma_t^X(p)} \in T_{\sigma_t^X(p)}\mathcal{M}$ and $Y_p \in T_p\mathcal{M}$. Therefore, the most intuitive way is to "pull it back", i.e., push it forward along the flow lines of X by negative time. ✎

Theorem 3.9 (Lie Bracket and Lie Derivative).

$$\mathcal{L}_X Y = [X, Y].$$

That is,

$$\left. \frac{d}{dt}(\sigma_{-t*}^X(Y_{\sigma_t^X(p)})) \right|_{t=0} = X \circ Y - Y \circ X.$$

Proof. We start with

$$\begin{aligned} \left. \frac{d}{dt}(\sigma_{-t*}^X(Y_{\sigma_t^X(p)})) \right|_{t=0} f &= \lim_{t \rightarrow 0} \frac{\sigma_{-t*}^X(Y_{\sigma_t^X(p)}f) - Y_p f}{t} \\ &= \lim_{t \rightarrow 0} \frac{Y_{\sigma_t^X(p)}(f \circ \sigma_{-t}^X) - Y_{\sigma_0^X(p)}(f \circ \sigma_0^X)}{t}. \end{aligned}$$

It motivates us to consider the function $H(r, s) = Y_{\sigma_r^X(p)}(f \circ \sigma_{-s}^X)$. Then we see

$$\begin{aligned} \left. \frac{d}{dt}(\sigma_{-t*}^X(Y_{\sigma_t^X(p)})) \right|_{t=0} f &= \left. \frac{d}{dt} H(t, t) \right|_{t=0} \\ &= \frac{\partial H}{\partial r}(0, 0) + \frac{\partial H}{\partial s}(0, 0). \end{aligned}$$

(r part: moving along X)

$$\begin{aligned} \frac{\partial H}{\partial r}(0, 0) &= \left. \frac{\partial}{\partial r} Y_{\sigma_r^X(p)}(f \circ \underbrace{\sigma_0^X}_{\text{id}}) \right|_{r=0} \\ &= \left. \frac{\partial}{\partial r} (Yf)(\sigma_r^X(p)) \right|_{r=0} \\ &= X_p(Yf). \end{aligned}$$

(s part: moving along Y)

$$\begin{aligned} \frac{\partial H}{\partial s}(0, 0) &= \left. \frac{\partial}{\partial s} Y_p(f \circ \sigma_{-s}^X) \right|_{s=0} \\ &= Y_p \left. \frac{\partial}{\partial s} (f \circ \sigma_{-s}^X) \right|_{s=0} \\ &= -Y_p(Xf), \end{aligned}$$

and therefore completes the proof. □

4 Formal Differential Form

✂ Remark.

In this section, we follow a local, coordinate approach. We focus on the requirements that makes form a form. We will postpone the realization of differential forms. □

4.1 Euclidean Spaces

Definition 4.1 (Formal Differential Form on $\mathbb{R}^m$).

A formal differential k -form on $\mathbb{R}^m$ is composed of m^k functions, arranged in the form

$$\omega = \sum_{1 \leq i_1, \dots, i_k \leq m} \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}.$$

We require,

1. All the functions $\omega_{i_1 \dots i_k}$ are $(-\infty, 0] \times \mathbb{R}^{m-1} \rightarrow (-\infty, 0] \times \mathbb{R}^{m-1}$ and C^∞ .
2. The operation $+$ is commutative and associative, just like the usual addition.
3. The operation $\wedge$ is associative and distributes over $+$, resembling the usual multiplication.
4. $\wedge$ also has anticommutativity,

$$\begin{cases} dx^i \wedge dx^j = -dx^j \wedge dx^i & \forall i \neq j \\ dx^i \wedge dx^i = 0 \end{cases}$$

Definition 4.2 (Equality of Formal Differential k -forms).

Let two formal k -forms ω, η be

$$\begin{aligned} \omega &= \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k} \\ \eta &= \eta_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k} \end{aligned}$$

We say they are "equal", denoted $\omega \equiv \eta$, iff

$$\sum_{\sigma \in S_k} \omega_{\sigma(i_1) \dots \sigma(i_k)} = \sum_{\sigma \in S_k} \eta_{\sigma(i_1) \dots \sigma(i_k)} \quad \forall 1 \leq i_1 < \dots < i_k \leq m.$$

✂ **Remark.** 1. For a formal differential k -form $\omega = dx^1 \wedge dx^2$, $\omega_{12} = 1$, but $\omega_{21} = 0$. The m^k components in the definition just presents a general form, so that it includes the scenario $\omega' = dx^1 \wedge dx^2 - dx^2 \wedge dx^1$. If you insist on arranging indices even in the definition, then ω' would not satisfy the definition, which is wierd.

2. For demonstration of equality, consider the following example in $\mathbb{R}^3$,

$$\begin{aligned} \omega &= dx^1 \wedge dx^2 + dx^2 \wedge dx^3 \\ \eta &= -dx^2 \wedge dx^1 + dx^2 \wedge dx^3. \end{aligned}$$

Choose $i_1 = 1, i_2 = 2$, and

$$S_3 = \{\underbrace{\text{id}, (1\ 2\ 3), (1\ 3\ 2)}_{\text{even}}, \underbrace{(1\ 2), (2\ 3), (1\ 3)}_{\text{odd}}\}.$$

Then, in order,

$$\begin{aligned} &\sum_{\sigma \in S_3} (\text{sgn } \sigma) \omega \\ &= \omega_{12} + \omega_{23} + \omega_{31} - \omega_{21} - \omega_{13} - \omega_{32} \\ &= 1 + 1 + 0 - 0 - 0 - 0. \end{aligned}$$

Notice how all permutations of a given component (paired in color) appears exactly once in this relation, and the sign is fixed accordingly by $\text{sgn } \sigma$.

3. Viewed this way, we see in the language of formal differential forms,

$$\begin{aligned} \omega &= dx^1 \wedge dx^2 \\ \eta &= -dx^2 \wedge dx^1 \\ \nu &= dx^1 \wedge dx^2 - dx^2 \wedge dx^1, \end{aligned}$$

$\omega \equiv \eta$, since $\omega_{12} - \omega_{21} = 1 = \eta_{12} - \eta_{21}$. But $\omega \not\equiv \nu$, since $\nu_{12} - \nu_{21} = 2$.

□

4.2 Operations

Definition 4.3 (Exterior Product).

Let two formal k -forms be $\omega = \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}$, $\eta = \eta_{j_1 \dots j_l} dx^{j_1} \wedge \dots \wedge dx^{j_l}$. Then

$$\omega \wedge \eta := \omega_{i_1 \dots i_k} \eta_{j_1 \dots j_l} dx^{i_1} \wedge \dots \wedge dx^{i_k} \wedge dx^{j_1} \wedge \dots \wedge dx^{j_l}.$$

Which is equivalent Definition 4.2 to,

$$\sum_{1 \leq s_1 < \dots < s_{k+l} \leq m} \left(\sum_S (\text{sgn } \sigma) \omega_{i_1 \dots i_k} \eta_{j_1 \dots j_l} \right) dx^{s_1} \wedge \dots \wedge dx^{s_{k+l}}.$$

Where S is all the combinations of $S_1 = \{i_1, \dots, i_k, j_1, \dots, j_l\}$ that is equal to $S_2 = \{s_1, \dots, s_{k+l}\}$, and the order does not matter. σ is the function $S_1 \rightarrow S_2$.

✍ Remark.

Example. Let $\dim \mathcal{M} = 6$, and

$$\begin{aligned} \omega &= \omega_{12} dx^1 \wedge dx^2 + \omega_{21} dx^2 \wedge dx^1 \\ \eta &= \eta_{456} dx^4 \wedge dx^5 \wedge dx^6. \end{aligned}$$

Then

$$\omega \wedge \eta = (\omega_{12} \eta_{456} - \omega_{21} \eta_{456}) dx^{1,2,4,5,6}$$

□

Definition 4.4 (Pullback).

Let $f : U \subseteq \mathbb{R}^m \rightarrow V \subseteq (-\infty, 0] \times \mathbb{R}^{n-1}$ is C^∞ . Choose local coordinates on U to be $x = (x^1, \dots, x^m)$, on V to be $y = (y^1, \dots, y^n)$. Define $f^\nu := y^\nu \circ f$.

Let $\omega = \omega_{j_1 \dots j_k} dy^{j_1} \wedge \dots \wedge dy^{j_k}$ be a formal k -form on U . Then

$$f^* \omega := (\omega_{j_1 \dots j_k} \circ f) \frac{\partial f^{j_1}}{\partial x^{i_1}} \dots \frac{\partial f^{j_k}}{\partial x^{i_k}} dx^{i_1} \wedge \dots \wedge dx^{i_k}.$$

- ✍ **Remark.** 1. The motivation is just somehow $\omega \circ f$. Chain the component function with f , and express the (resulting) coordinates as $dy^\nu = \frac{\partial f^\nu}{\partial x^\mu} dx^\mu$.
2. It is possible that $U \subseteq V$. Pulling back onto a subset is essentially a "limitation" on ω .
3. It is also possible that $U \subseteq \partial V$, i.e. U is the boundary of V . $f : p \mapsto (0, p)$ is just the immersion map in that case.

□

Definition 4.5 (Exterior Differentiation).

Let a formal k -form be $\omega = \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}$. Then

$$d\omega := \left(\frac{\partial \omega_{i_1 \dots i_k}}{\partial x^{i_0}} dx^{i_0} \right) \wedge dx^{i_1} \wedge \dots \wedge dx^{i_k}.$$

✍ **Remark.**

Let $U \subseteq \mathbb{R}^3$.

1. Consider a formal 0-form, i.e. $f \in C^\infty$. Then

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz \rightarrow \nabla f.$$

2. Consider a formal 1-form $\omega = Pdx + Qdy + Rdz$. Then

$$\begin{aligned} d\omega &= \left(\frac{\partial P}{\partial y} dy + \frac{\partial P}{\partial z} dz \right) \wedge dx + \left(\frac{\partial Q}{\partial x} dx + \frac{\partial Q}{\partial z} dz \right) \wedge dy \\ &\quad + \left(\frac{\partial R}{\partial x} dx + \frac{\partial R}{\partial y} dy \right) \wedge dz \\ &= \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) dy \wedge dz + \left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z} \right) dx \wedge dz \\ &\quad + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \wedge dy \\ &\rightarrow \text{curl } \omega. \end{aligned}$$

3. Consider a formal 2-form $\eta = A dy \wedge dz + B dz \wedge dx + C dx \wedge dy$. Then

$$\begin{aligned} d\eta &= \frac{\partial A}{\partial x} dx \wedge dy \wedge dz + \frac{\partial B}{\partial y} dy \wedge dz \wedge dx + \frac{\partial C}{\partial z} dz \wedge dx \wedge dy \\ &= \left(\frac{\partial A}{\partial x} + \frac{\partial B}{\partial y} + \frac{\partial C}{\partial z} \right) dx \wedge dy \wedge dz \\ &\rightarrow \operatorname{div} \eta. \end{aligned}$$

4. In usual vector calculus terms, we say ∇ produces a vector, div produces a scalar, and curl produces a vector. This r -form to $(m - r)$ -form correspondance is provided by the Hodge star operation.

□

4.3 Differential Form as Equivalence Classes

4.3.1 Operations are Well-Defined

Theorem 4.1.

If $\omega \equiv \omega'$, $\eta \equiv \eta'$ are formal k -forms on $\mathbb{R}^m$, then

$$\omega \wedge \eta \equiv \omega' \wedge \eta'.$$

Theorem 4.2.

If $\omega \equiv \omega'$ are formal k -forms on $\mathbb{R}^m$, and $f \in C^\infty(\mathbb{R}^m)$, then

$$f^* \omega \equiv f^* \omega'.$$

Theorem 4.3.

If $\omega \equiv \omega'$ are formal k -forms on $\mathbb{R}^m$, then

$$d\omega \equiv d\omega'.$$

4.3.2 Equivalence Classes

Definition 4.6 (Euclidean Differential k-form).

Denote the set of all formal k -forms on $\mathbb{R}^m$ be $A^k(\mathbb{R}^m)$. Then the set of all differential k -forms on $\mathbb{R}^m$ is defined to be

$$\Lambda^k(\mathbb{R}^m) := A^k(\mathbb{R}^m)/\equiv.$$

4.4 Properties of Operations

Theorem 4.4 (Properties of Exterior Product).

Let $\omega \in \Lambda^k(\mathbb{R}^m)$, $\eta \in \Lambda^l(\mathbb{R}^m)$. Then

$$(\omega_1 + \omega_2) \wedge \eta = \omega_1 \wedge \eta + \omega_2 \wedge \eta$$

$$\omega \wedge \eta = (-1)^{kl} \eta \wedge \omega.$$

Theorem 4.5 (Properties of Pullback).

Let $f \in C^\infty$, $\omega \in \Lambda^k(\mathbb{R}^m)$. Then

$$f^*(\omega_1 + \omega_2) = f^*\omega_1 + f^*\omega_2$$

$$f^*(\omega_1 \wedge \omega_2) = (f^*\omega_1) \wedge (f^*\omega_2)$$

$$g^*(f^*\eta) = (f \circ g)^*\eta.$$

Theorem 4.6 (Properties of Exterior Differentiation).

Let $f \in C^\infty$, $\omega \in \Lambda^k(\mathbb{R}^m)$. Then

$$f^*(d\omega) = d(f^*\omega)$$

$$d(\omega \wedge \eta) = (d\omega) \wedge \eta + (-1)^k \omega \wedge (d\eta)$$

$$d(d\omega) = 0.$$

4.5 Differential Forms on Manifolds

4.5.1 Requirements of Manifold Forms

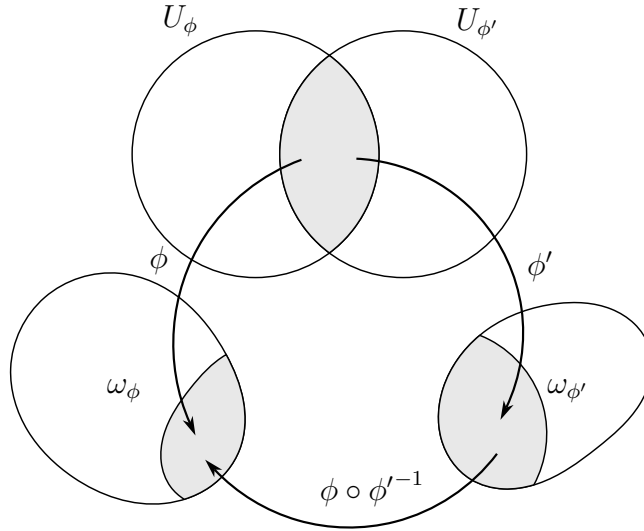
Definition 4.7 (Differential Form on a Manifold).

A C^∞ differential k -form ω on manifold $\mathcal{M}$, $\omega \in \Lambda^k(\mathcal{M})$, consists of a family of differential k -forms $\omega_\phi \in \Lambda^k(\phi(U_\phi))$, $\phi \in \Phi$, $\phi : U_\phi \rightarrow V_\phi = \phi(U_\phi) \subseteq (-\infty, 0] \times \mathbb{R}^{m-1}$, with an additional requirement that

$$\omega_{\phi'}|_{\phi'(U_\phi \cap U_{\phi'})} = \left(\phi \circ \phi'^{-1}\right)^* \left(\omega_\phi|_{\phi(U_\phi \cap U_{\phi'})}\right), \quad \forall \phi, \phi' \in \Phi.$$

ω_ϕ is called the local expression of ω on U_ϕ via ϕ .

✈ Remark.



The motivation is that, if two differential forms describe the same set, they should "agree" on that portion of manifold. The "agreement" is done by pull-back using the overlap function. ⌘

Theorem 4.7 (Covariance of Differential Forms).

Choose two overlapping charts $x^1, \dots, x^m$ and $y^1, \dots, y^m$ and define their Jacobian matrix [Definition 1.6](#) $J^\nu_\mu := \frac{\partial y^\nu}{\partial x^\mu} \big|_{\phi(p)}$, combining [Definition 4.4](#) and [Definition 4.7](#), we have

$$\begin{cases} dy^\nu = J^\nu_\mu dx^\mu & (\text{covariant}) \\ \omega_{\nu'} = \omega_\mu (J^{-1})^\mu_{\nu'} & (\text{contravariant}) \end{cases}$$

4.5.2 Operations on Manifold Forms**Definition 4.8 (Addition).**

Given two forms $\omega, \eta \in \Lambda^k(\mathcal{M})$, $m = \dim \mathcal{M}$, define their sum to be $\omega + \eta$, whose chart components are given by

$$(\omega + \eta)_\phi := \omega_\phi + \eta_\phi.$$

They satisfy [Definition 4.7](#) thanks to the linearity of pullback.

Definition 4.9 (Exterior Product).

Given two forms $\omega, \eta \in \Lambda^k(\mathcal{M})$, $m = \dim \mathcal{M}$, define their exterior product to be $\omega \wedge \eta$, whose chart components are given by

$$(\omega \wedge \eta)_\phi := \omega_\phi \wedge \eta_\phi.$$

They satisfy [Definition 4.7](#) because pullback commutes with exterior product.

Definition 4.10 (Exterior Differentiation).

Given a form $\omega \in \Lambda^k(\mathcal{M})$, $m = \dim \mathcal{M}$, define its exterior derivative to be $d\omega$, whose chart components are given by

$$(d\omega)_\phi := d(\omega_\phi).$$

They satisfy [Definition 4.7](#) because pullback commutes with exterior product.

Definition 4.11 (Pullback).

Given a form $\omega \in \Lambda^k(\mathcal{N})$ and a function $f : \mathcal{M} \rightarrow \mathcal{N}$, $m = \dim \mathcal{M}$, $n = \dim \mathcal{N}$. Choose coordinate functions $\phi_i : U_i \rightarrow \phi_i(U_i)$ on $\mathcal{M}$, and $\psi_j : V_j \rightarrow \psi_j(V_j)$ on $\mathcal{N}$.

To define $(f^*\omega)_{\phi_i}$, choose any $V_1, \dots, V_s$ s.t. $U_1 \subseteq \bigcup_{j=1}^s \tilde{f}(V_j)$. Then

$$(f^*\omega)_{\phi_i} := \sum_j (\phi_i \circ f \circ \psi_j^{-1})^* \left(\omega_{\psi_j} \big|_{\phi(f(U_i) \cap V_j)} \right)$$

They satisfy **Definition 4.7**.

5 Realization of Differential Forms

5.1 Cotangent Spaces

Definition 5.1 (Cotangent Spaces).

The cotangent space $T_p^*\mathcal{M}$ at $p \in \mathcal{M}$ is the set of all linear functions $f : T_p\mathcal{M} \rightarrow \mathbb{R}$.

Its member is called a cotangent vector.

$$\dim T_p^*\mathcal{M} = \dim T_p\mathcal{M}.$$

Definition 5.2 (One-Form).

A one-form on $\mathcal{M}$ is a smooth assignment of cotangent vectors $\omega : p \mapsto \omega_p$.

It may be understood as a covector field.

Definition 5.3 (Basis Cotangent Vectors).

The basis cotangent vectors is chosen to be the dual basis of the basis tangent vectors [Definition 2.8](#),

$$(dx^\mu)_p((\partial_\nu)_p) = \delta^\mu_\nu.$$

Theorem 5.1 (Coordinate Expression of Cotangent Vectors).

Any $f \in T_p^*\mathcal{M}$ can be expanded as

$$f = f_\mu(dx^\mu)_p.$$

Any one-form ω can be expressed as

$$\omega = \omega_\mu dx^\mu.$$

Theorem 5.2 (Pullback as Dual of Pushforward).

Given two manifolds $\mathcal{M}$ with dimension m and $\mathcal{N}$ with dimension n and a C^∞ function $f : \mathcal{M} \rightarrow \mathcal{N}$, the pullback of a one-form is the dual of pushforward. That is,

$$(h^*\omega)_p(v) := \omega_p(h_*v).$$

Theorem 5.3 (Local Representative of Pullback).

Let $\dim \mathcal{M} = m, \dim \mathcal{N} = n$, $h : \mathcal{M} \rightarrow \mathcal{N}$, $\{x^1, \dots, x^m\}$ be the local coordinates of $\mathcal{M}$ around p , and $\{y^1, \dots, y^n\}$ be the local coordinates of $\mathcal{N}$ around $h(p)$. Then

$$h^*\omega = \sum_{\mu=1}^m \sum_{\nu=1}^n \omega_\nu \left. \frac{\partial h^\nu}{\partial x^\mu} \right|_p (dx^\mu)_p,$$

where $J^\nu_\mu := \left. \frac{\partial h^\nu}{\partial x^\mu} \right|_p := (\partial_\mu)_p (y^\nu \circ h)$ is the Jacobian matrix.

Definition 5.4 (n-Forms).

An n -form is a tensor field of type $(0, n)$ that is totally skew-symmetric (or alternating, or totally antisymmetric), i.e.,

$$\omega(X_1, X_2, \dots, X_n) = (\text{sgn } \sigma) \omega(X_{\sigma(1)}, \dots, X_{\sigma(n)}), \quad \forall \sigma \in S_n.$$

The set of all n -forms on $\mathcal{M}$ is denoted as $\Lambda^n(\mathcal{M})$. The set of all forms is $\Lambda(\mathcal{M}) = \bigoplus_{n=0}^{\dim \mathcal{M}} \Lambda^n(\mathcal{M})$. Conventionally, we classify functions as 0-forms.

5.2 The Exterior Product

5.2.1 Definition and Properties

Definition 5.5 (Exterior Product).

Given $\omega_1 \in \Lambda^{n_1}(\mathcal{M})$, $\omega_2 \in \Lambda^{n_2}(\mathcal{M})$, their exterior product is a $(n_1 + n_2)$ -form given by,

$$\omega_1 \wedge \omega_2 := \frac{1}{n_1!n_2!} \sum_{\sigma \in S_{n_1+n_2}} (\text{sgn } \sigma) (\omega_1 \otimes \omega_2)_\sigma.$$

Written explicitly,

$$\begin{aligned} (\omega_1 \wedge \omega_2)(X_1, \dots, X_{n_1+n_2}) &:= \\ \frac{1}{n_1!n_2!} \sum_{\sigma \in S_{n_1+n_2}} (\text{sgn } \sigma) (\omega_1 \otimes \omega_2)(X_{\sigma(1)}, \dots, X_{\sigma(n_1+n_2)}) \end{aligned}$$

✍ Remark.

I'll take the alternating property and associativity of the exterior product for granted. For a detailed proof, see Hoffman. ✎

Theorem 5.4 (Commutativity with Pullback).

Given $h : \mathcal{M} \rightarrow \mathcal{N}$ and $\alpha, \beta \in \Lambda(\mathcal{N})$, then

$$h^*(\alpha \wedge \beta) = (h^*\alpha) \wedge (h^*\beta).$$

Remark.

For a "generalized" pullback, we have,

$$(h^*(\alpha))(X_1, \dots, X_{n_1}) = \alpha(h_*X_1, \dots, h_*X_{n_1}).$$

□

Proof.

$$\begin{aligned} & (h^*\alpha) \wedge (h^*\beta) \\ &= \frac{1}{n_1!n_2!} \sum_{\sigma \in S_{n_1+n_2}} (\text{sgn } \sigma) \alpha \otimes \beta(h_*X_{\sigma(1)}, \dots, h_*X_{\sigma(n_1+n_2)}). \\ &= \frac{1}{n_1!n_2!} \sum_{\sigma \in S_{n_1+n_2}} (\text{sgn } \sigma) h^*(\alpha \otimes \beta(X_{\sigma(1)}, \dots, X_{\sigma(n_1+n_2)})). \\ &= h^* \left(\frac{1}{n_1!n_2!} \sum_{\sigma \in S_{n_1+n_2}} (\text{sgn } \sigma) \alpha \otimes \beta(X_{\sigma(1)}, \dots, X_{\sigma(n_1+n_2)}) \right). \\ &= h^*(\alpha \wedge \beta). \end{aligned}$$

□

Theorem 5.5 (Skew-Symmetry).

The exterior product makes $\Lambda(\mathcal{M})$ a graded algebra with skew-symmetry given by

$$\omega_1 \wedge \omega_2 = (-1)^{n_1 n_2} \omega_2 \wedge \omega_1.$$

Proof. In the definition of exterior product, first fix $\sigma = \sigma_0$ to consider only one term.

When we switch ω_1 and ω_2 , we are essentially doing

$$\begin{aligned} & (\omega_2 \otimes \omega_1)(X_{\sigma_0(1)}, \dots, X_{\sigma_0(n_2)}, \underbrace{X_{\sigma_0(n_2+1)}, \dots, X_{\sigma_0(n_1+n_2)}}_{\text{...}}) \\ &= (\omega_1 \otimes \omega_2)(\underbrace{X_{\sigma_0(n_2+1)}, \dots, X_{\sigma_0(n_1+n_2)}}_{\text{...}}, X_{\sigma_0(1)}, \dots, X_{\sigma_0(n_2)}). \end{aligned}$$

Now,

$$\begin{aligned}
& \underbrace{1, 2, \dots, n_2}_{\downarrow \text{ } n_2 \text{ times}}, \underbrace{n_2 + 1, \dots, n_1 + n_2}_{\downarrow \text{ } (n_1 - 1)n_2 \text{ times}} \\
& \underbrace{n_2 + 1, \dots, n_1 + n_2}_{\downarrow \text{ } (n_1 - 1)n_2 \text{ times}}, \underbrace{1, 2, \dots, n_2}_{\downarrow \text{ } n_2 \text{ times}} \\
& \underbrace{n_2 + 1, \dots, n_1 + n_2}_{\downarrow \text{ } (n_1 - 1)n_2 \text{ times}}, \underbrace{1, 2, \dots, n_2}_{\downarrow \text{ } n_2 \text{ times}}
\end{aligned}$$

So $n_1 n_2$ transposes can achieve the desired effect. Therefore, every term in the summation is multiplied by $(-1)^{n_1 n_2}$, and we get the desired result. $\square$

Theorem 5.6 (Dimension of n-Forms).

Let $\dim \mathcal{M} = m$. If $1 \leq n \leq m$, then $\Lambda^n(\mathcal{M}) = \binom{m}{n}$. If $n > m$, then $\Lambda^n(\mathcal{M}) = 0$. Moreover, a basis for $\Lambda^n(\mathcal{M})_p$ is given by,

$$(dx^{\mu_1})_p \wedge (dx^{\mu_2})_p \wedge \dots \wedge (dx^{\mu_n})_p, \quad 1 \leq \mu_1 \leq \dots \leq \mu_n \leq m.$$

Remark.

The proof is quite a pleasure to read (and to think of). Please see Hoffman. $\square$

5.2.2 Evaluation

Theorem 5.7 (Evaluate k-forms with Vector Fields).

On a smooth manifold $\mathcal{M}$, under the chart $\phi = (x^1, \dots, x^m)$, let k vector fields have local representation $X_b = \xi^a_b \partial_a$. Then

$$(dx^1 \wedge \dots \wedge dx^k)(X_1, \dots, X_k) = \det \xi^a_b.$$

Proof.

$$\begin{aligned}
& (dx^1 \wedge \dots \wedge dx^k)(X_1, \dots, X_k) \\
&= (dx^1 \wedge \dots \wedge dx^k)(\xi^{i_1}_1 \partial_{i_1}, \dots, \xi^{i_k}_k \partial_{i_k}) \\
&= \xi^{i_1}_1 \xi^{i_k}_k (dx^1 \wedge \dots \wedge dx^k)(\partial_{i_1}, \dots, \partial_{i_k}).
\end{aligned}$$

To evaluate the last term, we first see that only terms with $i_\mu \neq i_\nu$ appear. Thus, we rewrite it using permutations,

$$\begin{aligned} & \xi^{i_1}_1 \xi^{i_k}_k (dx^1 \wedge \cdots \wedge dx^k)(\partial_{i_1}, \dots, \partial_{i_k}) \\ &= \sum_{\sigma \in S_k} \xi^{\sigma(1)}_1 \xi^{\sigma(k)}_k (dx^1 \wedge \cdots \wedge dx^k)(\partial_{\sigma(1)}, \dots, \partial_{\sigma(k)}). \end{aligned}$$

Moreover, since exterior product is alternating,

$$\begin{aligned} & \sum_{\sigma \in S_k} \xi^{\sigma(1)}_1 \xi^{\sigma(k)}_k (dx^1 \wedge \cdots \wedge dx^k)(\partial_{\sigma(1)}, \dots, \partial_{\sigma(k)}) \\ &= \sum_{\sigma \in S_k} (\text{sgn } \sigma) \xi^{\sigma(1)}_1 \cdots \xi^{\sigma(k)}_k \\ &= \det \xi^a_b. \end{aligned}$$

□

Corollary 5.7.1 (Evaluate k-forms with Vector Fields).

On a smooth manifold $\mathcal{M}$, let k covectors ω^i and k vector fields X_i , $1 \leq i \leq k$, and let the matrix $\xi^a_b = \omega^a(X_b)$. Then

$$(\omega^1 \wedge \cdots \wedge \omega^k)(X_1, \dots, X_k) = \det \xi^a_b.$$

Proof. This is easily proved if we use row-linearity of determinants, for $\omega^i = \omega^i_\mu dx^\mu$.

□

5.3 The Exterior Derivative

Definition 5.6 (Exterior Derivative).

Let a k -form be $\omega = \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}$. Then

$$d\omega := \left(\frac{\partial \omega_{i_1 \dots i_k}}{\partial x^{i_0}} dx^{i_0} \right) \wedge dx^{i_1} \wedge \dots \wedge dx^{i_k}.$$

If $\omega \in \Lambda^{\dim \mathcal{M}}(\mathcal{M})$, we define $d\omega = 0$.

Theorem 5.8.

A useful formula for the evaluation of the exterior derivative of a 0-form $f \in C^\infty(\mathcal{M})$ is the following,

$$df(X) := \mathcal{L}_X f = Xf.$$

Proof. In coordinates,

$$df = (\partial_\mu f)(dx^\mu).$$

So

$$df(X) = (\partial_\mu f)(dx^\mu)(X) = X^\mu (\partial_\mu f) = (X^\mu \partial_\mu) f = Xf.$$

□

Theorem 5.9 (Invariant Formula of Exterior Derivative for 1-forms).

In particular for a 1-form ω ,

$$d\omega(X, Y) = \mathcal{L}_X(\omega(Y)) - \mathcal{L}_Y(\omega(X)) - \omega([X, Y]).$$

Theorem 5.10 (Exterior Derivative and Product).

$$d(\omega_1 \wedge \omega_2) = d\omega_1 \wedge \omega_2 + (-1)^{\deg \omega_1} \omega_1 \wedge d\omega_2.$$

Theorem 5.11 (Exterior Derivative and Pullback).

Given $h : \mathcal{M} \rightarrow \mathcal{N}$, ω an n -form on $\mathcal{N}$, then

$$d(h^* \omega) = h^*(d\omega).$$

Theorem 5.12 (Functional Linearity of Exterior Derivative).

Let ω be a 1-form on $\mathcal{M}$. Then $d\omega$ satisfies,

$$d\omega(fX, Y) = f d\omega(X, Y), \quad \forall f \in C^\infty(\mathcal{M}),$$

where fX is a vector field that gives $(fX)(p) = f(p)X_p$.

Proof. By **Theorem 5.9**,

$$d\omega(fX, Y) = \mathcal{L}_{fX}(\omega(Y)) - \mathcal{L}_Y(\omega(fX)) - \omega([fX, Y]).$$

We break it down term by term. Firstly,

$$(\mathcal{L}_{fX}(\omega(Y)))(p) = f(p)X_p(\omega(Y)) = f(p)(\mathcal{L}_X(\omega(Y)))(p).$$

So

$$\mathcal{L}_{fX}(\omega(Y)) = f \cdot \mathcal{L}_X(\omega(Y)).$$

Secondly, we tackle $\mathcal{L}_Y(\omega(fX))$. In particular,

$$\omega(fX)(p) = \omega_p(f(p)X_p) = f(p)\omega_p(X_p) = f(p)(\omega(X))(p).$$

Therefore,

$$\mathcal{L}_Y(\omega(fX)) = \mathcal{L}_Y(f \cdot \omega(X)) = (\mathcal{L}_Y f)\omega(X) + f \cdot \mathcal{L}_Y(\omega(X)).$$

Thirdly,

$$\omega([fX, Y]) = \omega((fX) \circ Y - Y \circ (fX)).$$

In particular,

$$((Y \circ (fX))(g))(p) = Y_p((fX)(g)) = Y_p(f \cdot Xg) = (Y_p f)((Xg)(p)) + f(p) \cdot Y_p(Xg).$$

So,

$$Y \circ (fX) = (\mathcal{L}_Y f)X + f \cdot Y \circ X.$$

Substituting back,

$$\begin{aligned} \omega([fX, Y]) &= \omega(f \cdot X \circ Y - (\mathcal{L}_Y f)X - f \cdot Y \circ X) \\ &= \omega(f[X, Y] - (\mathcal{L}_Y f)X) \\ &= f\omega([X, Y]) - (\mathcal{L}_Y f)\omega(X). \end{aligned}$$

Finally,

$$\begin{aligned}
d\omega(fX, Y) &= \mathcal{L}_{fX}(\omega(Y)) - \mathcal{L}_Y(\omega(fX)) - \omega([fX, Y]) \\
&= f \cdot \mathcal{L}_X(\omega(Y)) - (\mathcal{L}_Y f)\omega(X) - f \cdot \mathcal{L}_Y(\omega(X)) - f\omega([X, Y]) + (\mathcal{L}_Y f)\omega(X) \\
&= f(\mathcal{L}_X(\omega(Y)) - \mathcal{L}_Y(\omega(X)) - \omega([X, Y])) \\
&= fd\omega(X, Y).
\end{aligned}$$

□

Corollary 5.12.1 (Local Nature of Exterior Derivative).

When ω is fixed, the value of $d\omega$ depends only on the local values of vector fields.

$$d\omega(X, Y)(p) = X^\mu(p)Y^\nu(p)d\omega(\partial_\mu, \partial_\nu)(p).$$

Proof. Write $X = X^\mu \partial_\mu$, noting that $X^\mu \in C^\infty(\mathcal{M})$, and use **Theorem 5.12**. □

5.4 DeRham Cohomology

Theorem 5.13 (Twice Exterior Differential).

For all $\omega \in \Lambda^n(\mathcal{M})$, $1 \leq n \leq \dim M$, we have

$$d^2\omega = 0.$$

✍ Remark.

This means

$$\text{Im}(d : \Lambda^{n-1}(\mathcal{M}) \rightarrow \Lambda^n(\mathcal{M})) \subseteq \text{Ker}(d : \Lambda^n(\mathcal{M}) \rightarrow \Lambda^{n+1}(\mathcal{M})).$$

This type of structure is called a differential complex, and is common in many structures. ✍

Definition 5.7 (Closed Form).

An n -form ω is closed if $d\omega = 0$. The set of all closed n -forms is denoted $Z^n(\mathcal{M})$.

Definition 5.8 (Exact Form).

An n -form ω is exact if $\omega = d\beta$ for some $(n-1)$ -form β . The set of all exact n -forms is denoted $B^n(\mathcal{M})$.

✍ Remark.

It is guaranteed that $B^n(\mathcal{M}) \subseteq Z^n(\mathcal{M})$, that is, exactness implies closure. But how much closed form is not exact is the study of cohomology theory. ✍

Theorem 5.14 (Poincare's Lemma).

On Euclidean space $\mathbb{R}^m$,

$$B^n(\mathcal{M}) = Z^n(\mathcal{M}), \quad \forall n > 0.$$

Definition 5.9 (DeRham Cohomology Groups).

The DeRham cohomology groups $H^n(\mathcal{M})$, $0 \leq n \leq \dim \mathcal{M}$ are the quotient

spaces

$$H^n(\mathcal{M}) := Z^n(\mathcal{M})/B^n(\mathcal{M}).$$

✍ **Remark.**

Recall the definition of quotient groups that $H^n(\mathcal{M})$ consists of elements of form $z + B^n(\mathcal{M})$, $z \in Z^n(\mathcal{M})$.

If all closed forms are exact, $Z^n(\mathcal{M}) \subseteq B^n(\mathcal{M})$, then $H^n(\mathcal{M}) \simeq \{0\}$. □

Theorem 5.15 (Criterion of Exact ODE).

On the Euclidean space $\mathbb{R}^2$, given a 1-form $\omega = \omega_1 dx^1 + \omega_2 dx^2$. Then

$$\omega \in B^1(\mathbb{R}^2) \iff \partial_2 \omega_1 = \partial_1 \omega_2.$$

✍ **Remark.**

This is an important theorem to me, for it connects the "exactness of differential forms" to the familiar notion of "exactness of differential equations".

It also provides the first hints that we are actually integrating forms, and that exterior differentiation of a 0-form resembles gradient in usual vector calculus terms. □

Proof. Via Poincare lemma [Theorem 5.14](#), on $\mathbb{R}^2$, exactness is equivalent to closure. So we need only to determine the condition that $d\omega = 0$. Using [Definition 5.6](#),

$$\begin{aligned} d\omega &= \partial_\nu \omega_{\mu_1} dx^\nu \wedge dx^{\mu_1} \\ &= \partial_2 \omega_1 dx^2 \wedge dx^1 + \partial_1 \omega_2 dx^1 \wedge dx^2 \\ &= (\partial_2 \omega_1 - \partial_1 \omega_2) dx^2 \wedge dx^1. \end{aligned}$$

□

6 Tensors

Definition 6.1 (Tensors).

If $\dim \mathcal{M} \neq \infty$, the tensors of type (r, s) $T_p^{r,s} \mathcal{M}$ are all the linear functions

$$f : \bigtimes_{p=1}^r T_p^* \mathcal{M} \times \bigtimes_{p=1}^s T_p \mathcal{M} \rightarrow \mathbb{R}.$$

I.e., it eats r covectors and s vectors.

Theorem 6.1 (Dimensions of General Tensor Space).

The dimension of $T_p^{r,s} \mathcal{M}$ is $m^r m^s$. In particular, a basis for the space is,

$$\bigotimes_{1 \leq \mu_1 \dots \mu_r \leq m} (\partial_{\mu_i})_p \otimes \bigotimes_{1 \leq \nu_1 \dots \nu_s \leq m} (dx^{\nu_i})_p$$

Remark.

For a detailed proof, see Hoffman.

□

Theorem 6.2 (Transformation Properties of Tensor Fields).

Given a manifold $\mathcal{M}$ of dimension m , choose two overlapping charts $\phi : U \rightarrow V$ with local coordinates $x^1, \dots, x^m$, $\phi' : U' \rightarrow V'$ with $x'^1, \dots, x'^m$. A tensor field T of type (r, s) with local representation on U given by

$$T^{\mu_1 \dots \mu_r}_{\nu_1 \dots \nu_s} \partial_{\mu_1} \otimes \dots \otimes \partial_{\mu_r} \otimes dx^{\nu_1} \otimes \dots \otimes dx^{\nu_s}$$

transforms like

$$T^{\mu'_1 \dots \mu'_r}_{\nu'_1 \dots \nu'_s} = T^{\mu_1 \dots \mu_r}_{\nu_1 \dots \nu_s} \frac{\partial x'^{\mu'_1}}{\partial x^{\mu_1}} \dots \frac{\partial x'^{\mu'_r}}{\partial x^{\mu_r}} \frac{\partial x^{\nu_1}}{\partial x'^{\nu'_1}} \dots \frac{\partial x^{\nu_s}}{\partial x'^{\nu'_s}}$$

Definition 6.2 (Pullback of Covariant Tensor).

The pullback of a general tensor field τ by $f : \mathcal{M} \rightarrow \mathcal{N}$ is defined as

$$(f^* \tau)(X_1, \dots, X_k) := (\tau \circ f)(f_* X_1, \dots, f_* X_k).$$

6.1 The Metric Tensor

6.1.1 Riemannian Metric

Definition 6.3 (Riemannian Metric).

A Riemannian metric on a manifold $\mathcal{M}$ is a totally symmetric, smooth $(2,0)$ -tensor field on $\mathcal{M}$ that is positive definite at every point $p \in \mathcal{M}$. Locally,

$$g = g_{ij} dx^i \otimes dx^j.$$

Definition 6.4 (The Euclidean Metric).

On the Euclidean space $\mathbb{R}^m$, there is a natural metric called the Euclidean metric, which is locally defined as

$$\bar{g} = \delta_{ij} dx^i \otimes dx^j.$$

Definition 6.5 (The Product Metric).

Given two Riemannian manifolds $(\mathcal{M}, g_{\mathcal{M}}), (\mathcal{N}, g_{\mathcal{N}})$, we can form a metric $\hat{g}$ on the product manifold $\mathcal{M} \times \mathcal{N}$ by

$$\hat{g}((v_1, v_2), (w_1, w_2)) = g_{\mathcal{M}}(v_1, w_1) + g_{\mathcal{N}}(v_2, w_2).$$

6.1.2 Existence of Riemannian Metric

Theorem 6.3 (Existence of Riemannian Metric).

Every smooth manifold admits a Riemannian metric [Definition 6.3](#).

Proof. Let the manifold have coordinate charts $\phi_j, j \in J$. Let a smooth partition of unity $\rho_j, j \in J$ subordinate to U_j . Then we can form a smooth tensor field

$$g = \sum_j \rho_j \phi_j^* g_j,$$

where g_j is the Euclidean metric on V_j . It is obviously symmetric. Since the partition of unity is positive and every g_j is positive definite, g is positive definite also. $\square$

6.1.3 Pullback Metric

Definition 6.6 (The Pullback Metric).

Let $f : \mathcal{M} \rightarrow \mathcal{N}$ be C^∞ , where $\mathcal{N}$ is a Riemannian manifold equipped with Riemannian metric g . Then f^*g is a Riemannian metric on $\mathcal{M}$ iff f is a smooth immersion.

Theorem 6.4 (A Useful Mnemonic for the Pullback Metric).

Let $f : \mathcal{M} \rightarrow \mathcal{N}$ be a smooth immersion so that Definition 6.6 can be defined. Then

$$(f^*g) = (g_{ij} \circ f) d(y_i \circ f)^2 \otimes d(y_j \circ f)^2,$$

where $d(y_i \circ f)$ expands like $\frac{\partial(y_i \circ f)}{\partial x^j} dx^j$, just like the exterior derivative.

Proof.

$$\begin{aligned} f^*g(X_1, X_2) &= (g \circ f)(f_*X_1, f_*X_2) \\ &= (g_{ij} \circ f) dy^i(f_*X_1) dy^j(f_*X_2) \\ &= (g_{ij} \circ f) dy^i(\partial_{\nu_1} \frac{\partial f^{\nu_1}}{\partial x^{\mu_1}} X_1^{\mu_1}) dy^j(\partial_{\nu_2} \frac{\partial f^{\nu_2}}{\partial x^{\mu_2}} X_2^{\mu_2}) \\ &= (g_{ij} \circ f) \frac{\partial f^i}{\partial x^{\mu_1}} \frac{\partial f^j}{\partial x^{\mu_2}} X_1^{\mu_1} X_2^{\mu_2}. \end{aligned}$$

So

$$(f^*g)_{\mu\nu} = (g_{ij} \circ f) \frac{\partial f^i}{\partial x^{\mu_1}} \frac{\partial f^j}{\partial x^{\mu_2}}.$$

But

$$d(y^i \circ f) := \frac{\partial(y^i \circ f)}{\partial x^j} dx^j.$$

Tensor product all those completes the proof. □

6.1.4 Musical Isomorphism

Definition 6.7 (Musical Isomorphism).

$$\begin{aligned} (X^\flat)_\mu &= g_{\mu\nu} X^\nu \\ (\omega^\sharp)^\mu &= g^{\mu\nu} \omega_\nu, \end{aligned}$$

where $g^{\mu\nu} := g_{\mu\nu}^{-1}$.

7 Integration of Differential Forms

7.1 Partition of Unity

Definition 7.1 (Support).

Let X be a topological space, and $f : X \rightarrow \mathbb{R}$. Then the support of f is defined as

$$\text{supp } f := \{x \in X \mid f(x) \neq 0\}.$$

Theorem 7.1 (Partition of Unity).

Let $\mathcal{M}$ be a C^∞ manifold with dimension m with atlas Φ .

Let $\Phi = \{\phi_j \mid \phi_j : V_j \rightarrow \phi_j(V_j), j \in J\}$.

Then it is possible to construct a set of C^∞ functions $\rho_j, j \in J$ s.t.

$$1 = \sum_{j \in J} \rho_j, \text{ supp } \rho_j \subseteq V_j$$

7.2 Orientation

7.2.1 Definition

Definition 7.2 (Compatible Coordinate Charts).

Given a manifold $\mathcal{M}$, its two coordinate charts are called compatible (have the same orientation) if,

$$\det J > 0.$$

Where J is the Jacobian matrix [Definition 1.6](#).

If the manifold has a maximal compatible atlas, then we say the manifold is orientable, and we may call its corresponding orientation positive and denote the atlas Φ_+ .

Theorem 7.2.

A manifold has either no orientation (any atlas is not compatible) or two orientations.

Theorem 7.3 (Orientability and Existence of Forms of Highest Degree).

A manifold is orientable iff there exists a nowhere vanishing differential form of the highest degree.

7.2.2 Positively Oriented Boundary**Definition 7.3 (Positively Oriented Boundary).**

Let $\mathcal{M}$ be a orientable C^∞ manifold with dimension m , positively oriented by compatible atlas Φ_+ . Define coordinate charts on $\partial\mathcal{M}$ from Φ as follows,

$$\phi^{\partial\mathcal{M}} : U_\phi \cap \partial\mathcal{M} \rightarrow \mathbb{R}^{m-1},$$

Then $\Phi_+^{\partial\mathcal{M}} := \{\phi^{\partial\mathcal{M}} \mid \phi \in \Phi_+\}$ determines an orientation on $\partial\mathcal{M}$, called the positive orientation.

7.3 Pseudoforms (Densities)**7.3.1 Pseudoscalars****Definition 7.4 (Space of Pseudoscalars).**

Given a manifold $\mathcal{M}$ and a point p on it. Choose an arbitrary orientation $o_p = \Phi_+$ of $T_p\mathcal{M}$. On the set $\mathbb{R} \times \{o_p, -o_p\}$, define an equivalence relation,

$$\equiv : (c, o_p) \equiv (-c, -o_p).$$

We denote $\tilde{P}_p := \mathbb{R} \times \{o_p, -o_p\} / \equiv$, and call it the space of pseudoscalars. Under a choice of local orientation o , we often denote its element $(c, o) \equiv (-c, -o)$ by $\tilde{c}_o$.

Theorem 7.4 (Pseudoscalar Vector Space).

The space $\tilde{P}_p$ forms a vector space under the operations,

$$\begin{aligned}(a, o) + (b, o) &= (a + b, o), \\ c(a, o) &= (ca, o).\end{aligned}$$

Specifically, $\dim \tilde{P}_p = 1$, since $\forall \tilde{c}_o \in \tilde{P}_p, \exists c \in \mathbb{R}$ s.t. $\tilde{c}_o = c\tilde{1}_o$.

Theorem 7.5 (Pseudoscalar Vector Bundle).

On the space $\tilde{P} = \bigcup_{p \in \mathcal{M}} \tilde{P}_p$, define the canonical projection $\pi : \tilde{P} \rightarrow \mathcal{M}$ by $\tilde{P}_p \mapsto p$.

For any chart $\phi : U \rightarrow \mathbb{R}^m \in \Phi$ of the underlying manifold $\mathcal{M}$, it induces an orientation on $T_p\mathcal{M}$, denoted by o_ϕ . We define the local trivialization by

$$\begin{aligned}\tilde{\phi} : \tilde{\pi}(U) &\rightarrow U \times \mathbb{R} \\ (p, (c, o_\phi)) &\mapsto (p, c).\end{aligned}$$

And this makes pseudoscalars a vector bundle on any manifold $\mathcal{M}$.

Theorem 7.6 (Pseudoscalars Under Coordinate Transformation).

Given two charts $\phi : U \rightarrow \mathbb{R}^m, \psi : V \rightarrow \mathbb{R}^m \in \Phi$ of a manifold $\mathcal{M}$ and a point $p \in U \cap V$, the following relationship holds on the pseudoscalar space $\tilde{P}_p$

Definition 7.4,

$$\tilde{c}_\phi = (\text{sgn det } J)\tilde{c}_\psi,$$

where $\tilde{c}_\phi$ means using the orientation induced by ϕ , and $\text{sgn det } J$ is the sign of the Jacobian matrix **Definition 1.6**.

Proof. If $o_\phi = o_\psi$, that is, $\det J > 0$ and their orientations agree, then $(c, o_\phi) \equiv (c, o_\psi)$. But if $o_\phi = -o_\psi$, that is, $\det J < 0$ and their orientations disagree, then $(c, o_\phi) \equiv (-c, o_\psi)$.

This shows that pseudoscalars "flip sign" under charts of different orientation. $\square$

7.3.2 Definition

Definition 7.5 (Pseudoforms).

The space of pseudoforms $\tilde{\Lambda}^k(\mathcal{M})$ is defined by the space of differential forms tensored with (twisted by) the pseudoscalar bundle,

$$\tilde{\Lambda}^k(\mathcal{M}) = \bigwedge^k T^*\mathcal{M} \otimes \tilde{P}(\mathcal{M}).$$

This means, pseudoforms are defined locally as

$$\tilde{\omega} = \tilde{1}_\phi \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}.$$

7.3.3 Pseudoforms and Forms on Orientable Manifolds

Theorem 7.7 (Conversion of Pseudoforms and Forms).

For a pseudoform $\tilde{\omega} \in \tilde{\Lambda}^k(\mathcal{M})$ on an orientable manifold $\mathcal{M}$ of degree m positively oriented by the atlas Φ_+ , the positive orientation is chosen in a continuous manner. Then $\tilde{P}(\mathcal{M})$ is isomorphic to $\mathbb{R}$ by the positive orientation in a continuous manner, so we can say

$$\begin{aligned} \iota : \tilde{\Lambda}^k(\mathcal{M}) &\rightarrow \Lambda^k(\mathcal{M}) \\ \tilde{1}_\phi \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k} &\mapsto o(\phi) \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k} \end{aligned}$$

defines a smooth differential form on $\mathcal{M}$. Moreover, we can also convert it back,

$$\begin{aligned} \iota^{-1} : \Lambda^k(\mathcal{M}) &\rightarrow \tilde{\Lambda}^k(\mathcal{M}) \\ \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k} &\mapsto \tilde{1}_\phi o(\phi) \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}. \end{aligned}$$

7.3.4 On Absolute Values

✂ Remark.

Some people would say for $m = \dim \mathcal{M}$, pseudo- m -forms are absolute values of m -forms. This literal definition is doable in our formulation, and is given by the following theorem. □

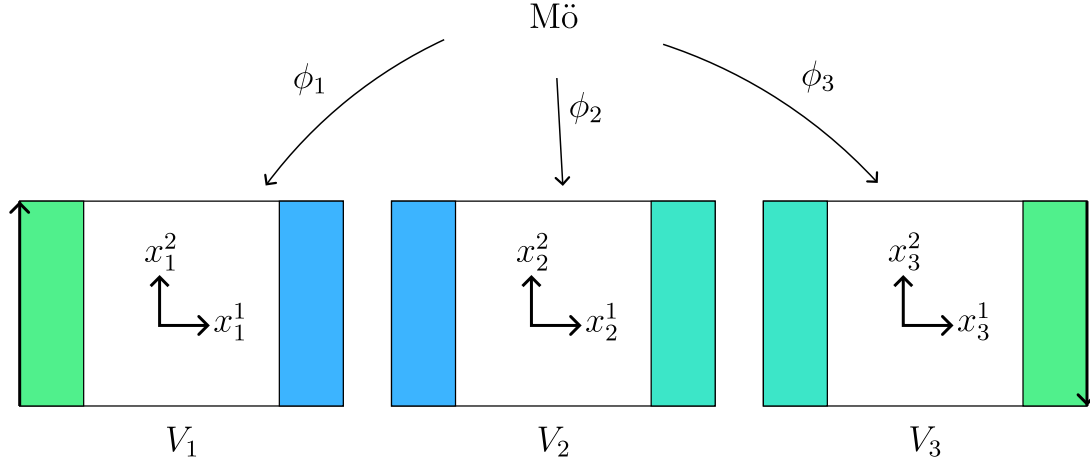
Theorem 7.8.

Given vectors $v_1, \dots, v_m \in T_p \mathcal{M}$, denote their orientation by o_v . Let $\tilde{o}_v(\tilde{1}_\phi) : \tilde{P}_p \rightarrow \mathbb{R}$ by $(c, o_v) \mapsto c$.

$$\begin{aligned} & (\tilde{o}_v(\tilde{1}_\phi) f(x) dx^1 \wedge \dots \wedge dx^m) (v_1, \dots, v_m) \\ &= |(f(x) dx^1 \wedge \dots \wedge dx^m) (v_1, \dots, v_m)|. \end{aligned}$$

✈ **Remark.**

To understand these in a deeper level, we should think about why we cannot have a nowhere-vanishing differential form on a non-orientable manifold. Let's try to define a usual form on the Möbius band.



We create a Möbius band by patching three charts together, with overlapping areas indicated by the same color. $V_3 \cap V_1$ are patched together twisted, indicated by the reverse direction of arrows.

Let's say we have the three forms

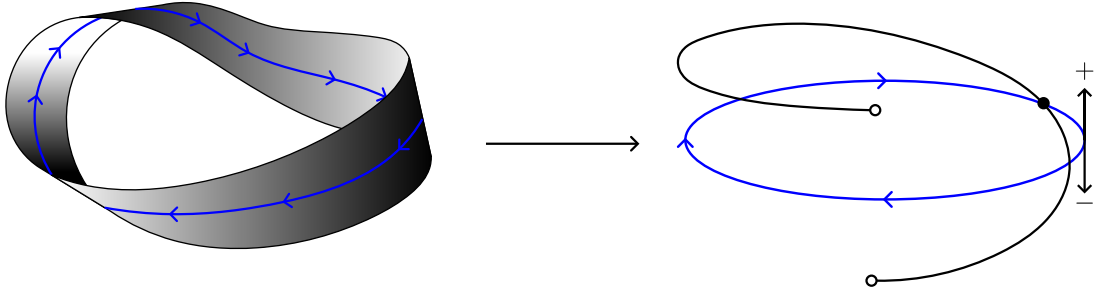
$$\begin{aligned} \omega_1 &= dx_1^1 \wedge dx_1^2 \\ \omega_2 &= f_2 dx_2^1 \wedge dx_2^2 \\ \omega_3 &= f_3 dx_3^1 \wedge dx_3^2 \end{aligned}$$

defined on V_1, V_2, V_3 , respectively, and with ω_1 prescribed to be 1 everywhere.

Then by the pullback criterion, **Definition 4.7**, we have

$$\begin{aligned}
\omega_1 &= (\phi_2 \circ \phi_1^{-1})^* \omega_2 \\
&= f_2(\phi_2 \circ \phi_1^{-1}(x_1)) dx_1^1 \wedge dx_1^2 \\
\omega_2 &= (\phi_3 \circ \phi_2^{-1})^* \omega_3 \\
&= f_3(\phi_3 \circ \phi_2^{-1}(x_2)) dx_2^1 \wedge dx_2^2 \\
\omega_3 &= (\phi_1 \circ \phi_3^{-1})^* \omega_1 \\
&= -f_1(\phi_1 \circ \phi_3^{-1}(x_1)) dx_3^1 \wedge dx_3^2.
\end{aligned}$$

But this cannot be, since by evaluating backwards from the last equation, we have $f_3 = -1$, $f_2 = f_3 = -1$, and $f_1 = f_2 = -1$. This contradicts with the given fact that $f_1 = 1$. By this reasoning, we have found an example of the fact that one cannot have a nowhere-vanishing usual form on a non-orientable manifold. Schematically, it looks like this,



where the black line resembles the value of ω . As you can see, it has a mismatch of sign, making it discontinuous. It is possible to resolve this situation by taking the absolute value, but if so, the linearity is broken. Introducing the pseudoscalar allows us to say $\tilde{1}_\phi = -\tilde{1}_\psi$ if their orientations are reversed, so that we can sort of "fold" the ends together, making it continuous again. In terms of coordinates,

$$\begin{aligned}
\tilde{\omega}_3 &= (\phi_1 \circ \phi_3^{-1})^* \tilde{\omega}_1 \\
&= -\tilde{1}_{\phi_1} f_1(\phi_1 \circ \phi_3^{-1}(x_3)) dx_3^1 \wedge dx_3^2 \\
&= \tilde{1}_{\phi_3} f_1(\phi_1 \circ \phi_3^{-1}(x_3)) dx_3^1 \wedge dx_3^2 \\
&= \tilde{1}_{\phi_3} dx_3^1 \wedge dx_3^2.
\end{aligned}$$

□

7.3.5 Metric Volume Form

Definition 7.6 (The Metric Volume Form).

On a smooth Riemannian manifold $\mathcal{M}$ of dimension m with metric g , the metric volume form is the unique pseudo- m -form that gives $\tilde{1}_\phi$ if the vectors are orthonormal and fits the local orientation of ϕ . Especially in 3-dimensional spaces, we denote it as vol .

Theorem 7.9 (Local Representation of Metric Volume Form).

On a smooth Riemannian manifold $\mathcal{M}$ of dimension m with metric g and coordinate chart $\phi = x^1, \dots, x^m$, the metric volume form has local representation

$$\text{vol} = \tilde{1}_\phi \sqrt{\det g} \, dx^1 \wedge \dots \wedge dx^m,$$

where $\det g$ means $\det g_{ij}$, the determinant of the matrix of metric tensor.

Proof. Let $\text{vol} = \tilde{1}_\phi dx^1 \wedge \dots \wedge dx^m$ with the tangent vectors $(\partial_1)_p, \dots, (\partial_m)_p$ orthonormal in the sense of g . Then $g_{\mu\nu} = \delta_{\mu\nu}$ in that basis.

Under another coordinates $\psi = (y^1, \dots, y^m)$, we have

$$\begin{aligned} \text{vol} &= \tilde{1}_\phi \det \left(\frac{\partial x}{\partial y} \right) dy^1 \wedge \dots \wedge dy^m \\ &= \tilde{1}_\psi \left| \det \left(\frac{\partial x}{\partial y} \right) \right| dy^1 \wedge \dots \wedge dy^m. \end{aligned}$$

But

$$\begin{aligned} g &= \delta_{ij} dx^i \otimes dx^j \\ &= \delta_{ij} \frac{\partial x^i}{\partial y^\alpha} \frac{\partial x^j}{\partial y^\beta} dy^\alpha \otimes dy^\beta \end{aligned}$$

by Theorem 6.2. So

$$\begin{aligned} g_{\alpha\beta} &= \sum_k \frac{\partial x^k}{\partial y^\alpha} \frac{\partial x^k}{\partial y^\beta} \\ &= \sum_k (J^{-1})_{k\alpha} (J^{-1})_{k\beta} \\ &= \sum_k ((J^{-1})^\top)_{\alpha k} (J^{-1})_{k\beta} \\ &= ((J^{-1})^\top J^{-1})_{\alpha\beta}. \end{aligned}$$

Thus

$$\det g_{\alpha\beta} = \det ((J^{-1})^\top J^{-1}) = (\det(J^{-1}))^2 = \left| \det \frac{\partial x}{\partial y} \right|^2$$

and the proof is completed. $\square$

7.4 Integration of Forms of Highest Degree

Definition 7.7 (Integration).

Let $\mathcal{M}$ be a paracompact C^∞ manifold of dimension m . Choose a C^∞ partition of unity $\rho_j, j \in J$ of $\mathcal{M}$ s.t. $\text{supp } \rho_j \subseteq U_{\phi_j} := U_j$.

Let a pseudo- m -form $\tilde{\omega} \in \tilde{\Lambda}^m(\mathcal{M})$ has local expression $\tilde{\omega}_{\phi_j} = \tilde{1}_{\phi_j} f_j(x) dx_j^1 \wedge \cdots \wedge dx_j^m$.

$$\int_{\mathcal{M}} \tilde{\omega} = \sum_{j \in J} \int_{(-\infty, 0] \times \mathbb{R}^{m-1}} (\rho_j \circ \phi_j^{-1})(x) f_j(x) dx^1 \dots dx^m$$

if the finite sum exists and has the same value for all choices of ρ_j and ϕ_j .

Remark.

The following theorem reveals why we integrate pseudoforms, not usual forms.

$\mathfrak{M}$

Theorem 7.10 (Criterion of Existence of Integral).

If $\text{supp } \tilde{\omega}$ is compact, then $\int_{\mathcal{M}} \tilde{\omega}$ exists.

Proof. Let two sets of coordinate charts be

$$\begin{aligned} \phi_j &: U_j \rightarrow V_j, j \in J \\ \phi'_k &: U'_k \rightarrow V'_k, k \in K. \end{aligned}$$

And cooresponding partition of unity be ρ_j, ρ'_k .

(The goal) Show

$$\begin{aligned} & \sum_{j \in J} \int (\rho_j \circ \phi_j^{-1})(x) f_j(x) dx^1 \dots dx^m \\ &= \sum_{k \in K} \int (\rho'_k \circ \phi'_k{}^{-1})(x') f'_k(x') dx'^1 \dots dx'^m \end{aligned}$$

(Split using ρ'_k)

$$\begin{aligned}\int_{\mathcal{M}} \tilde{\omega} &= \sum_{j \in J} \int (\rho_j \circ \phi_j^{-1})(x) f_j(x) dx^1 \dots dx^m \\ &= \sum_{j \in J} \int \sum_{k \in K} (\rho'_k \circ \phi_j^{-1})(x) (\rho_j \circ \phi_j^{-1})(x) f_j(x) dx^1 \dots dx^m.\end{aligned}$$

Since the sum is finite, and $\text{supp } \tilde{\omega}$ is compact, and therefore the integral is not improper; thus, there can be no limit or Fubini problems on exchanging sums and integrals. So

$$\int_{\mathcal{M}} \tilde{\omega} = \sum_{j \in J} \sum_{k \in K} \int (\rho_j \rho'_k \circ \phi_j^{-1})(x) f_j(x) dx^1 \dots dx^m$$

(Change of variables) First fix j, k .

$$\begin{aligned}& \int (\rho_j \rho'_k \circ \phi_j^{-1})(x) f_j(x) dx^1 \dots dx^m \\ &= \int (\rho_j \rho'_k \circ \phi_j^{-1})(\phi_j \circ \phi'_k{}^{-1}(x')) f_j(\phi_j \circ \phi'_k{}^{-1}(x')) \left| \det \left(\frac{\partial x}{\partial x'} \right) \right| dx'^1 \dots dx'^m \\ &= \int (\rho_j \rho'_k \circ \phi'_k{}^{-1}(x')) f_j(\phi_j \circ \phi'_k{}^{-1}(x')) \left| \det \left(\frac{\partial x}{\partial x'} \right) \right| dx'^1 \dots dx'^m\end{aligned}$$

(Pullback Condition) From **Definition 4.7**,

$$\tilde{\omega}_{\phi'_k} = \left(\phi_j \circ \phi'_k{}^{-1} \right)^* \tilde{\omega}_{\phi_j}.$$

Using **Theorem 7.6**, we see,

$$\begin{aligned}& \tilde{\mathbf{1}}_{\phi'_k} f'_k(x') dx'^1 \wedge \dots \wedge dx'^m \\ &= \tilde{\mathbf{1}}_{\phi_j} f_j(\phi_j \circ \phi'_k{}^{-1}(x')) \left(\frac{\partial x^1}{\partial x'^{\ell_1}} dx'^{\ell_1} \right) \wedge \dots \wedge \left(\frac{\partial x^m}{\partial x'^{\ell_m}} dx'^{\ell_m} \right) \\ &= \tilde{\mathbf{1}}_{\phi_j} f_j(\phi_j \circ \phi'_k{}^{-1}(x')) \sum_{\sigma \in S_m} (\text{sgn } \sigma) \frac{\partial x^1}{\partial x'^{\sigma(1)}} \dots \frac{\partial x^m}{\partial x'^{\sigma(m)}} dx'^1 \wedge \dots \wedge dx'^m \\ &= (\text{sgn } \det J) \tilde{\mathbf{1}}_{\phi'_k} f_j(\phi_j \circ \phi'_k{}^{-1}(x')) \det \left(\frac{\partial x}{\partial x'} \right) dx'^1 \wedge \dots \wedge dx'^m \\ &= \tilde{\mathbf{1}}_{\phi'_k} f_j(\phi_j \circ \phi'_k{}^{-1}(x')) \left| \det \left(\frac{\partial x}{\partial x'} \right) \right| dx'^1 \wedge \dots \wedge dx'^m.\end{aligned}$$

So we see

$$f'_k(x') = f_j(\phi_j \circ \phi'_k{}^{-1}(x')) \left| \det \left(\frac{\partial x}{\partial x'} \right) \right|.$$

Therefore, the integral

$$\begin{aligned} & \int (\rho_j \rho'_k \circ \phi'_k{}^{-1}(x')) f_j(\phi_j \circ \phi'_k{}^{-1}(x')) \left| \det \left(\frac{\partial x}{\partial x'} \right) \right| dx'^1 \dots dx'^m \\ &= \int (\rho_j \rho'_k \circ \phi'_k{}^{-1}(x')) f'_k(x') dx'^1 \dots dx'^m \end{aligned}$$

(Closing) By moving the sum wrt $j \in J$ into the integral and using the property of partition of unity, the proof is completed. $\square$

7.5 Integration of Forms of Lower Degree

7.5.1 Pseudoorientation (Transverse Orientation)

Remark.

Because pseudoscalars depend on local orientations, we need an extra structure in order to pullback pseudoforms onto a submanifold. $\mathfrak{M}$

Definition 7.8 (Pseudoorientation).

For $\mathcal{Z}$ a submanifold of $\mathcal{M}$, pseudoorientation is a map that for all $p \in \mathcal{Z}$, takes a local orientation of $\mathcal{Z}$ at p (an orientation of $T_p\mathcal{Z}$) to a local orientation of $\mathcal{M}$ at p (an orientation of $T_p\mathcal{M}$) **continuously in p , and takes opposite orientations to opposite orientations.**

Remark.

Requiring well-defined surface normal (with a metric, of course) is precisely this criterion for $(m-1)$ dimensional submanifolds, for we can form an orientation of $\mathcal{M}$ from an orientation of $\mathcal{Z}$ (which is essentially $(m-1)$ vectors on $T_p\mathcal{Z}$) and the normal.

This makes Mö integrable as a submanifold of $\text{Mö} \times \mathbb{R}$, since we can choose the surface normal to be positive everywhere. But Mö is not integrable as a submanifold of $\mathbb{R}^3$, since if we try to assign surface normal along the strip, when we get back to where we started, the direction would be reversed. $\mathfrak{M}$

Definition 7.9 (Pullback of Pseudoscalars).

Given a manifold $\mathcal{M}$ and its submanifold $\mathcal{Z}$, and ψ be a chart of $\mathcal{M}$, ϕ a chart of $\mathcal{Z}$. At the point p , if the pseudoorientation i **Definition 7.8** is chosen such that the local orientations $o_\phi \mapsto o_\psi$, then we say $i^* \tilde{1}_\psi = \tilde{1}_\phi$. If $o_\phi \mapsto -o_\psi$, then $i^* \tilde{1}_\psi = -\tilde{1}_\phi$.

7.5.2 Definition**Definition 7.10 (Integration of Lower Degree Forms).**

Let $\mathcal{Z}$ be an pseudooriented C^∞ submanifold of dimension d of $\mathcal{M}$, $i : \mathcal{Z} \rightarrow \mathcal{M}$ be the immersion map.

Let $\omega \in \Lambda^d(\mathcal{M})$, we define

$$\int_{\mathcal{Z}} \omega := \int_{\mathcal{Z}} i^* \omega$$

if it exists.

7.5.3 Stoke's theorem**Theorem 7.11 (Stoke's Theorem).**

If $\mathcal{M}$ is an oriented C^∞ manifold of dimension m and $\omega \in \Lambda^{m-1}(\mathcal{M})$, then

$$\int_{\mathcal{M}} d\omega = \int_{\partial\mathcal{M}} \omega := \int_{\partial\mathcal{M}} i^* \omega,$$

where $i : \partial\mathcal{M} \rightarrow \mathcal{M}$ is just the immersion map, $i : p \mapsto p$.

Proof. (Partition Using Charts) Choose a C^∞ partition of unity **Theorem 7.1** $\rho_j, j \in J$ s.t. $\text{supp } \rho_j$ are compact and $\text{supp } \rho_j \subseteq U_{\phi_j} := U_j$.

Now $\omega = \sum_{j \in J} \rho_j \omega$ is a finite sum. So it suffices to show that, if $\eta \in \Lambda^{m-1}(\mathcal{M})$, $\text{supp } \eta$ compact and $\text{supp } \eta \subseteq U_\phi$ then $\int_{\mathcal{M}} d\eta = \int_{\partial\mathcal{M}} \eta$, and apply $\eta = \rho_j \omega$ for all $j \in J$.

(The integral) Suppose the coordinates of ϕ is labeled $x^1, \dots, x^m$. Locally, let

$$\eta = \sum_{\ell=1}^m f_\ell dx^1 \wedge \cdots \cancel{dx^\ell} \cdots \wedge dx^m,$$

where $f_\ell : (-\infty, 0] \times \mathbb{R}^{m-1} \rightarrow \mathbb{R}$, and it is C^∞ . Then, also locally,

$$\begin{aligned} d\eta &= \sum_{\ell=1}^m \frac{\partial f_\ell}{\partial x^\ell} dx^\ell \wedge dx^1 \wedge \cdots \cancel{dx^\ell} \cdots \wedge dx^m \\ &= \left(\sum_{\ell=1}^m (-1)^{\ell-1} \frac{\partial f_\ell}{\partial x^\ell} \right) dx^1 \wedge \cdots \wedge dx^m. \end{aligned}$$

Then by **Definition 7.7**,

$$\int_{\mathcal{M}} d\eta = \int_{(-\infty, 0] \times \mathbb{R}^{m-1}} \left(\sum_{\ell=1}^m (-1)^{\ell-1} \frac{\partial f_\ell}{\partial x^\ell} \right) dx^1 \cdots dx^m.$$

(To be precise, we need to choose another partition of unity ρ'_j to do integration. But we can just choose it to cover all of $\text{supp } \eta$ and don't care all other parts, so that doesn't matter too much.)

Choose a rectangular region R s.t.

$$\text{supp } \eta \subseteq [a_1, 0] \times \cdots \times [a_m, b_m]$$

and define

$$R_\ell := [a_1, 0] \times \cdots \cancel{[a_\ell, b_\ell]} \cdots \times [a_m, b_m].$$

($\ell = 2, \dots, m$) In this case, by Fubini and FTC,

$$\begin{aligned} & \int_R \frac{\partial f_\ell}{\partial x^\ell} dx^1 \cdots dx^m \\ &= \int_{R_\ell} \left(\int_{a_\ell}^{b_\ell} \frac{\partial f_\ell}{\partial x^\ell} dx^\ell \right) dx^1 \cdots \cancel{dx^\ell} \cdots dx^m \\ &= \int_{R_\ell} \underbrace{(f_\ell(x^1, \dots, b_\ell, \dots, x^m))}_0 - \underbrace{(f_\ell(x^1, \dots, a_\ell, \dots, x^m))}_0 \cancel{dx^\ell} \cdots dx^m \\ &= 0, \end{aligned}$$

since $\text{supp } \eta \subseteq R$, so on the boundary $f = 0$.

($\ell = 1$) Now the integral has only one term left.

$$\begin{aligned}
\int_{\mathcal{M}} d\eta &= \int_R \frac{\partial f_1}{\partial x^1} dx^1 \dots dx^m \\
&= \int_{R_1} \left(\int_{a_1}^0 \frac{\partial f_1}{\partial x^1} dx^1 \right) dx^2 \dots dx^m \\
&= \int_{R_\ell} (f_1(0, x^2, \dots, x^m) - \underbrace{f_1(a_1, x^2, \dots, x^m)}_0) dx^2 \dots dx^m \\
&= \int_{\mathbb{R}^{m-1}} (f_1 \circ i) dx^2 \dots dx^m \\
&= \int_{\partial \mathcal{M}} i^* \eta.
\end{aligned}$$

□

7.5.4 Integrate Without Partition of Unity

✍ Remark.

The truth is, no one can (or wants) to write down a partition of unity. If one do write it down, integrating it is a big problem.

Fortunately, if one decompose any manifold $\mathcal{M}$ into finitely many submanifolds $\mathcal{M}_i, i \in I$ which intersect only on their boundaries, then $\int_{\mathcal{M}} \tilde{\omega} = \sum_{i \in I} \int_{\mathcal{M}_i} f_i^* \tilde{\omega}$, where f_i is the injective immersion map $f_i : \mathcal{M}_i \rightarrow \mathcal{M}$. ☞

7.6 Hodge Dual

7.6.1 The Dual Inner Product

Theorem 7.12 (Uniqueness of Dual Inner Product).

Let $\mathcal{M}$ be an m -dimensional Riemannian manifold with metric g . Then for all $k = 1, \dots, m$, g determines a unique inner product $\langle \cdot, \cdot \rangle$ s.t.

$$\langle \omega^1 \wedge \dots \wedge \omega^k, \eta^1 \wedge \dots \wedge \eta^k \rangle = \det \left(\left\langle (\omega^i)^\sharp, (\eta^j)^\sharp \right\rangle_g \right)$$

Proof. (Show dx^I is an orthonormal basis) Let I, J be increasing k -indices, and let $\partial_1, \dots, \partial_k$ be local orthonormal frame. Thus, $g_{ij} = \delta_{ij}$, and $(dx^i)^\sharp = \partial_i$.

If $I = J$,

$$\langle dx^I, dx^I \rangle = \det(\langle \partial_i, \partial_j \rangle) = 1.$$

But if $I \neq J$, $\langle dx^I, dx^J \rangle = 0$ since there would be zero rows.

Now define inner product by

$$\langle \omega, \eta \rangle = \langle \omega_I dx^I, \eta_J dx^J \rangle := \omega_I \eta_J \delta^{IJ}.$$

(I cannot show it satisfy the desired property; but I convinced myself using examples.)

(It remains to show that any other orthonormal frame produces the same inner product. But I hadn't think of a good way.) $\square$

Theorem 7.13 (Volume Form Is Of Unit Norm).

On a Riemannian manifold $\mathcal{M}$ with dimension m and metric g , the metric volume form vol is the only m -form that has unit norm with respect to the inner product **Theorem 7.12**.

Proof. Let an m -form $\omega = \lambda dx^1 \wedge \cdots \wedge dx^m$. We wish

$$\begin{aligned} 1 &= \langle \omega, \omega \rangle = \lambda^2 \langle dx^1 \wedge \cdots \wedge dx^m, dx^1 \wedge \cdots \wedge dx^m \rangle \\ &= \lambda^2 \det \left(\left\langle (dx^i)^\sharp, (dx^j)^\sharp \right\rangle \right) \\ &= \lambda^2 \det \left(g_{\mu\nu} (dx^i)^\sharp{}^\mu (dx^j)^\sharp{}^\nu \right). \end{aligned}$$

Now, since $(dx^i)_\mu = \delta^i_\mu$, we know $(dx^i)^\sharp{}^\mu = g^{\mu\nu} \delta^i_\nu = g^{\mu i}$. Moreover, $g_{\mu\nu} g^{\nu\sigma} = \delta_\mu^\sigma$. So

$$\begin{aligned} 1 &= \lambda^2 \det (g_{\mu\nu} g^{\mu i} g^{\nu j}) \\ &= \lambda^2 \det (g^{i\mu} g_{\mu\nu} g^{\nu j}) \\ &= \lambda^2 \det (g^{i\mu} \delta_\mu^j) \\ &= \lambda^2 \det (g^{ij}). \end{aligned}$$

Therefore, $\lambda = \pm \sqrt{\det (g_{ij})}$. $\square$

7.6.2 Definition and Uniqueness

Theorem 7.14 (Defining Property of Hodge Star).

On a Riemannian manifold $\mathcal{M}$ with dimension m and metric g , there exists a unique bundle homomorphism $\star : \Lambda^k(\mathcal{M}) \rightarrow \tilde{\Lambda}^{m-k}(\mathcal{M})$ s.t.

$$\omega \wedge \star \eta = \langle \omega, \eta \rangle_g \text{vol}.$$

Proof. (uniqueness) Suppose there exist two such homomorphisms, $\star$ and $\star'$. Then for all $\omega \in \Lambda^k(\mathcal{M})$,

$$\omega \wedge \star \eta = \langle \omega, \eta \rangle_g \text{vol}$$

$$\omega \wedge \star' \eta = \langle \omega, \eta \rangle_g \text{vol}$$

so $\omega \wedge (\star \eta - \star' \eta) = 0$. Thus $\star \eta = \star' \eta$ for all $\eta \in \Lambda^k(\mathcal{M})$.

(existence) We claim for a local orthonormal frame ∂_i ,

$$\star (dx^{i_1} \wedge \cdots \wedge dx^{i_k}) = \epsilon^{i_1 \dots i_k}_{j_{k+1} \dots j_m} dx^{j_{k+1}} \wedge \cdots \wedge dx^{j_m}.$$

Since

$$\begin{aligned} & (dx^{i_1} \wedge \cdots \wedge dx^{i_k}) \wedge \star (dx^{i_1} \wedge \cdots \wedge dx^{i_k}) \\ &= \langle dx^I, dx^I \rangle \text{vol} \\ &= \text{vol}, \end{aligned}$$

we need the Levi-Civita symbol to make the indices in correct order. □

7.6.3 Coordinate Representation of Hodge Dual

Theorem 7.15 (Hodge Dual of Any Basis).

On a Riemannian manifold $\mathcal{M}$ with dimension m and metric g , the Hodge dual satisfies

$$\begin{aligned} & \star (dx^{i_1} \wedge \cdots \wedge dx^{i_k}) \\ &= \frac{\sqrt{\det g}}{(n-k)!} g^{i_1 j_1} \cdots g^{i_k j_k} \epsilon_{j_1 \dots j_k j_{k+1} \dots j_m} dx^{j_{k+1}} \wedge \cdots \wedge dx^{j_m}. \end{aligned}$$

The term $(n-k)!$ is there because we didn't restrict $j_{k+1}, \dots, j_m$ are in order. If we ensure that, then $(n-k)!$ is not there.

Proof. We verify it by calculating

$$(dx^{i_1} \wedge \cdots \wedge dx^{i_k}) \wedge \left(\frac{\sqrt{\det g}}{(n-k)!} g^{i_1 j_1} \cdots g^{i_k j_k} \epsilon_{j_1 \dots j_k j_{k+1} \dots j_m} dx^{j_{k+1}} \wedge \cdots \wedge dx^{j_m} \right).$$

Firstly,

$$\begin{aligned}
\langle dx^{i_1} \wedge \cdots \wedge dx^{i_k}, dx^{i_1} \wedge \cdots \wedge dx^{i_k} \rangle &= \det \left(\left\langle (dx^i)^\sharp, (dx^j)^\sharp \right\rangle \right) \\
&= \det (g_{\mu\nu} g^{i\mu} g^{j\nu}) \\
&= \det (g^{i\mu} g_{\mu\nu} g^{\nu j}) \\
&= \det (g^{ij}),
\end{aligned}$$

but only $i_1, \dots, i_k$ rows and columns are picked out.

However, $g^{i_1 j_1} \dots g^{i_k j_k} \epsilon_{j_1 \dots j_m}$ are precisely this expression, because $j_{k+1}, \dots, j_m$ cannot contain terms of $i_1, \dots, i_k$, otherwise they wedge to be 0. Therefore, $j_1, \dots, j_k$ are permutations of $i_1, \dots, i_k$, and the Levi-Civita symbol takes care of the $(\text{sgn } \sigma)$. Thus the proof is completed. $\square$

Corollary 7.15.1 (Coordinate Representation of Hodge Dual).

Given a k -form $\omega \in \Lambda^k(\mathcal{M})$ on a Riemannian manifold $\mathcal{M}$ with dimension m and metric g , we have

$$(\star\omega)_{j_{k+1} \dots j_m} = \frac{\sqrt{\det g}}{(n-k)!} g^{i_1 j_1} \dots g^{i_k j_k} \epsilon_{j_1 \dots j_k j_{k+1} \dots j_m} \omega_{i_1 \dots i_k}.$$

7.6.4 Properties of Hodge Dual

Theorem 7.16 (Double Hodge Dual).

For any $\omega \in \Lambda^k(\mathcal{M})$ on a Riemannian manifold $\mathcal{M}$,

$$\star\star\omega = (-1)^{k(m-k)}\omega.$$

✂ Remark.

Note this applies to Euclidean metric only. In other cases, the negative sign may be different. For example, under the Lorentzian metric $g = (-+++)$, the expression is instead

$$\star\star\omega = (-1)^{k(m-k)+1}\omega.$$

Also, be very careful about the k when dealing with consecutive Hodges. $\mathfrak{M}$

Proof. It suffices to show for orthonormal frames,

$$\star\star(dx^{i_1} \wedge \cdots \wedge dx^{i_k}) = (-1)^{k(n-k)}(dx^{i_1} \wedge \cdots \wedge dx^{i_k}).$$

To show this, we verify according to [Theorem 7.14](#),

$$\star (dx^{i_1} \wedge \cdots \wedge dx^{i_k}) \wedge (-1)^{k(n-k)} (dx^{i_1} \wedge \cdots \wedge dx^{i_k}).$$

We want this to be equal to $\langle \star dx^I, \star dx^I \rangle \text{vol}$. By the exchanging rule [Theorem 5.5](#), we have

$$\begin{aligned} &= (dx^{i_1} \wedge \cdots \wedge dx^{i_k}) \wedge \star (dx^{i_1} \wedge \cdots \wedge dx^{i_k}) \\ &= \langle dx^I, dx^I \rangle \text{vol}. \end{aligned}$$

Thus, if $\langle dx^I, dx^I \rangle = \langle \star dx^I, \star dx^I \rangle$, then we are done. I can prove the case when they are orthonormal,

$$\begin{aligned} &\langle \star (dx^{i_1} \wedge \cdots \wedge dx^{i_k}), \star (dx^{i_1} \wedge \cdots \wedge dx^{i_k}) \rangle \\ &= \left\langle \delta^{i_1 j_1} \cdots \delta^{i_k j_k} \epsilon_{j_1 \dots j_n} dx^{j_{k+1}} \wedge \cdots \wedge dx^{j_m}, \delta^{i_1 j'_1} \cdots \delta^{i_k j'_k} \epsilon_{j'_1 \dots j'_n} dx^{j'_{k+1}} \wedge \cdots \wedge dx^{j'_m} \right\rangle \\ &= 1 \end{aligned}$$

because only the terms which $j_{k+1}, \dots, j_m$ didn't contain $i_1, \dots, i_k$ survive, and Levi-Civita symbol squared is definitely 1. $\square$

Theorem 7.17 (Hodge Dual of Constant).

On any Riemannian manifold $\mathcal{M}$,

$$\star 1 = \text{vol}.$$

More generally, for any function $f \in C^\infty(\mathcal{M})$,

$$\star f = f \text{vol}.$$

Theorem 7.18 (Hodge Dual of Flatted Vector Fields).

On any Riemannian manifold $\mathcal{M}$, given a vector field X ,

$$\star X^\flat = X \lrcorner \text{vol}.$$

Proof. Again, we verify [Theorem 7.14](#). In local coordinates,

$$\begin{aligned} \text{vol} &= \sqrt{\det g} dx^1 \wedge \cdots \wedge dx^m \\ X &= X^\mu \partial_\mu \\ X^\flat &= (X^\flat)_\mu dx^\mu = g_{\mu\nu} X^\nu dx^\mu. \end{aligned}$$

Then, using [Theorem 8.2](#),

$$\begin{aligned} & (X^\flat) \wedge (X \lrcorner \text{vol}) \\ &= g_{\mu\nu} X^\nu \sqrt{\det g} \sum_{i=1}^m (-1)^{i-1} X^i dx^\mu \wedge dx^1 \wedge \dots \wedge \cancel{dx^i} \wedge \dots \wedge dx^m. \end{aligned}$$

But the terms that $\mu \neq i$ vanishes, so

$$\begin{aligned} & (X^\flat) \wedge (X \lrcorner \text{vol}) \\ &= \sqrt{\det g} \sum_{i=1}^m (-1)^{i-1} g_{i\nu} X^\nu X^i dx^1 \wedge \dots \wedge \cancel{dx^i} \wedge \dots \wedge dx^m \\ &= g_{\mu\nu} X^\mu X^\nu \text{vol} \\ &= \langle X, X \rangle_g \text{vol}. \end{aligned}$$

□

7.6.5 Other Differential Operators

Definition 7.11 (Codifferential).

On a Riemannian manifold $\mathcal{M}$ equipped with metric g , we define the codifferential $\delta : \Lambda^k(\mathcal{M}) \rightarrow \Lambda^{k-1}(\mathcal{M})$.

$$\begin{aligned} \delta &:= (-1)^k \star^{-1} d \star \\ &= (-1)^{m(k+1)+1} \star d \star. \end{aligned}$$

The second expression is given by [Theorem 7.16](#), since the inverse Hodge operator is itself times some negative sign.

Theorem 7.19 (Twice Codifferential).

$$\delta^2 = 0.$$

Proof.

$$\begin{aligned} \delta^2 &= (-1)^{k-1} \star^{-1} d \star (-1)^k \star^{-1} d \star \\ &= -\star^{-1} d^2 \star = 0. \end{aligned}$$

□

Theorem 7.20 (Codifferential Of Functions).

For any $f \in C^\infty(\mathcal{M})$,

$$\delta f = 0.$$

Proof.

$$\delta f = (-1)^0 \star^{-1} \underbrace{d \star f}_{f \text{ vol}}^0 = 0$$

□

Definition 7.12 (Divergence).

On a Riemannian manifold $\mathcal{M}$ equipped with metric g , we define the divergence of a vector field to be

$$\begin{aligned} \operatorname{div} X &:= \star d \star X^\flat \\ &= -\delta X^\flat. \end{aligned}$$

The second expression is given by [Definition 7.11](#) and the fact that $k = 1$.

Theorem 7.21.

The divergence of a vector field $\operatorname{div} X$ is the unique scalar that

$$(\operatorname{div} X) \operatorname{vol} = d \star X^\flat.$$

Proof. Note that

$$\operatorname{div} X = -\delta X^\flat = \star^{-1} d \star X^\flat.$$

Therefore,

$$\operatorname{div} X \wedge \star \operatorname{div} X = \langle \operatorname{div} X, \operatorname{div} X \rangle \operatorname{vol}.$$

But since $\operatorname{div} X$ is just a scalar, the above is equivalent to

$$(\operatorname{div} X) \star \operatorname{div} X = (\operatorname{div} X)^2 \operatorname{vol}.$$

If $\operatorname{div} X \neq 0$, then

$$\star \operatorname{div} X = d \star X^\flat = (\operatorname{div} X) \operatorname{vol}.$$

If $\operatorname{div} X = 0$, then by definition $d \star X^\flat = 0$, so the equation trivially holds. □

Theorem 7.22 (Divergence In Local Coordinates).

On a Riemannian manifold $\mathcal{M}$ equipped with metric g , the divergence **Definition 7.12** of a vector field X has local expression

$$\operatorname{div} X = \frac{1}{\sqrt{\det g}} \partial_\mu \left(X^\mu \sqrt{\det g} \right).$$

Proof. Using **Theorem 7.18** and **Theorem 8.2**,

$$\begin{aligned} \operatorname{div} X &= \star d \star X^\flat = \star d(X \lrcorner \operatorname{vol}) \\ &= \star d \left(\sum_{i=1}^m (-1)^{i-1} \sqrt{\det g} X^i dx^i \wedge \dots \cancel{dx^i} \dots \wedge dx^m \right) \\ &= \star \sum_{i=1}^m (-1)^{i-1} \partial_\mu \left(\sqrt{\det g} X^i \right) dx^\mu \wedge dx^i \wedge \dots \cancel{dx^i} \dots \wedge dx^m. \end{aligned}$$

Now only the $\mu = i$ terms remain, and since $\star \star 1 = (-1)^{0 \cdot (m-0)} = 1$, $\star \operatorname{vol} = 1$, and so

$$\begin{aligned} \operatorname{div} X &= \star \left(\partial_\nu \left(\sqrt{\det g} X^\nu \right) \frac{1}{\sqrt{\det g}} \operatorname{vol} \right) \\ &= \frac{1}{\sqrt{\det g}} \partial_\nu \left(\sqrt{\det g} X^\nu \right). \end{aligned}$$

□

Definition 7.13 (Laplace-de Rham Operator).

On a Riemannian manifold $\mathcal{M}$ equipped with metric g , we define the Laplace-de Rham operator to be

$$\Delta := -(d\delta + \delta d),$$

where δ is the codifferential **Definition 7.11**.

✍ Remark.

Both sign conventions are popular. This particular convention makes the local expression in Euclidean space with the standard metric g reduce to the familiar "Laplacian". In Minkowski space, the LdR operator is the wave operator □, the d'Alembertian. ✎

Theorem 7.23 (LdR Operator In Local Coordinates).

On a Riemannian manifold $\mathcal{M}$ equipped with metric g , the LdR operator **Definition 7.13** of a function f has local expression

$$\begin{aligned}\Delta f &= -\delta df = \operatorname{div} \operatorname{grad} f \\ &= \frac{1}{\sqrt{\det g}} \partial_\mu \left(\sqrt{\det g} g^{\mu\nu} \partial_\nu f \right).\end{aligned}$$

7.7 Integration of Vector Fields

7.7.1 Line Integrals

7.7.2 Surface Integrals

✍ **Remark.**

In vector calculus, we define the integration of a vector field X over a submanifold S to be

$$\int_{\mathcal{M}} \langle X, N \rangle_g dS,$$

where N is the normal vector of S , $\langle X, N \rangle_g$ is the inner product of the two (induced by metric g), and dS is the (metric) volume form of S . We seek to reformulate this in terms of our language. □

Theorem 7.24 (Hodge Star and Flux).

Let $\mathcal{M}$ be an m -dimensional Riemannian manifold with metric g , and S be a submanifold of $\mathcal{M}$ of codimension 1. Let X be a vector field on $\mathcal{M}$, dS be the volume forms of $\mathcal{M}$ and S , respectively, and i be the immersion map. Then

$$i^*(\star X^\flat) = \langle X, N \rangle_g dS.$$

Proof. Let $X^\perp = \langle X, N \rangle_g N$, $X^\top = X - X^\perp$. Then

$$X \lrcorner dV = X^\perp \lrcorner dV + X^\top \lrcorner dV.$$

But $X^\perp = \langle X, N \rangle_g N$, so

$$X^\perp \lrcorner dV = \langle X, N \rangle_g N \lrcorner dV = \langle X, N \rangle_g dS.$$

We now show $X^\top \lrcorner dV = 0$. Consider any $X_1, \dots, X_{m-1}$ on $\mathcal{S}$, we see

$$(X^\top \lrcorner dV)(X_1, \dots, X_{m-1}) = dV(X^\top, X_1, \dots, X_{m-1}) = 0$$

since m vectors on $(m-1)$ -dimensional vector spaces are necessarily dependent. $\square$

Theorem 7.25 (Divergence Theorem).

Let $\mathcal{M}$ be an m -dimensional Riemannian manifold with metric g . Denote the metric volume form of $\partial\mathcal{M}$ by dS , then for any compactly supported vector field X , we have

$$\int_{\mathcal{M}} (\operatorname{div} X) \operatorname{vol} = \int_{\partial\mathcal{M}} \langle X, N \rangle_g dS.$$

Proof. By [Theorem 7.21](#), and Stokes theorem [Theorem 7.11](#), we see

$$\int_{\mathcal{M}} (\operatorname{div} X) \operatorname{vol} = \int_{\mathcal{M}} d\star X^\flat = \int_{\partial\mathcal{M}} i^* (\star X^\flat) = \int_{\partial\mathcal{M}} \langle X, N \rangle_g dS.$$

$\square$

8 Lie Derivative of Forms and Derivative of Integrals

8.1 Interior Product (Contraction)

Definition 8.1 (Interior Product).

On a smooth manifold $\mathcal{M}$ of dimension m , let $v \in T_p\mathcal{M}$, $\alpha \in \Lambda^k(\mathcal{M})$. Then we define the interior product of the vector and the form to be

$$v \lrcorner \alpha = \begin{cases} 0 & \forall \alpha \in \Lambda^0(\mathcal{M}) \\ \alpha(v) & \forall \alpha \in \Lambda^1(\mathcal{M}) \\ \alpha(v, \dots) & \forall \alpha \in \Lambda^k(\mathcal{M}). \end{cases}$$

Theorem 8.1 (Basic Properties of Interior Product).

$$\begin{aligned} (v + cw) \lrcorner \alpha &= v \lrcorner \alpha + c(w \lrcorner \alpha) \\ v \lrcorner (\alpha + f\beta) &= v \lrcorner \alpha + f \cdot (v \lrcorner \beta) \end{aligned}$$

Theorem 8.2 (Interior Product and Cofactor Expansion).

For any $v \in T_p\mathcal{M}$ and k covectors $\omega^i \in \Lambda^1(\mathcal{M})$, $1 \leq i \leq k$, we have

$$v \lrcorner (\omega^1 \wedge \dots \wedge \omega^k) = \sum_{i=1}^k (-1)^{i-1} \omega^i(v) \omega^1 \wedge \dots \wedge \widehat{\omega^i} \wedge \dots \wedge \omega^k.$$

Proof. Form the matrix $\xi_j^i = \omega^i(v_j)$ as in [Corollary 5.7.1](#). We see

$$(\omega^1 \wedge \dots \wedge \omega^k)(v_1, \dots, v_k) = \det(\xi_j^i).$$

Cofactor expand this on the first column, and the theorem is proved. $\square$

Corollary 8.2.1 (Interior Product as an Antiderivation).

For any $v \in T_p\mathcal{M}$, $\alpha \in \Lambda^{n_1}(\mathcal{M})$, $\beta \in \Lambda^{n_2}(\mathcal{M})$, we have

$$v \lrcorner (\alpha \wedge \beta) = (v \lrcorner \alpha) \wedge \beta + (-1)^{n_1} \alpha \wedge (v \lrcorner \beta).$$

Proof. Let $\alpha = \alpha_{i_1 \dots i_{n_1}} dx^{i_1} \wedge \dots \wedge dx^{i_{n_1}}$, $\beta = \beta_{j_1 \dots j_{n_2}} dx^{j_1} \wedge \dots \wedge dx^{j_{n_2}}$. Then

$$\begin{aligned} v \lrcorner (\alpha \wedge \beta) &= v \lrcorner (\alpha_{i_1 \dots i_{n_1}} \beta_{j_1 \dots j_{n_2}} dx^{i_1} \wedge \dots \wedge dx^{i_{n_1}} \wedge dx^{j_1} \wedge \dots \wedge dx^{j_{n_2}}) \\ &= \alpha_{i_1 \dots i_{n_1}} \beta_{j_1 \dots j_{n_2}} (v \lrcorner (dx^{i_1} \wedge \dots \wedge dx^{i_{n_1}} \wedge dx^{j_1} \wedge \dots \wedge dx^{j_{n_2}})). \end{aligned}$$

But

$$\begin{aligned} (v \lrcorner \alpha) \wedge \beta &= \alpha_{i_1 \dots i_{n_1}} \beta_{j_1 \dots j_{n_2}} v \lrcorner (dx^{i_1} \wedge \dots \wedge dx^{i_{n_1}}) \wedge dx^{j_1} \wedge \dots \wedge dx^{j_{n_2}} \\ \alpha \wedge (v \lrcorner \beta) &= \alpha_{i_1 \dots i_{n_1}} \beta_{j_1 \dots j_{n_2}} dx^{i_1} \wedge \dots \wedge dx^{i_{n_1}} \wedge v \lrcorner (dx^{j_1} \wedge \dots \wedge dx^{j_{n_2}}). \end{aligned}$$

If we expand all these using [Theorem 8.2](#), we see that we need $\alpha \wedge (v \lrcorner \beta)$ to have a factor of $(-1)^{n_1}$, since we are essentially piecing two cofactors together. $\square$

Corollary 8.2.2 (Coordinate Representation).

$$(v \lrcorner \alpha)_{i_1 \dots i_k} = v^{i_0} \alpha_{i_0 i_1 \dots i_k}.$$

8.2 Lie Derivative of Forms

Definition 8.2 (Lie Derivative of Forms).

Given a form $\omega \in \Lambda^k(\mathcal{M})$ and a vector field X , we define the Lie derivative of ω by X to be a k -form,

$$(\mathcal{L}_X \omega)_p = \frac{d}{dt} (\sigma_t^{X*} \omega)_p \Big|_{t=0} = \lim_{t \rightarrow 0} \frac{\sigma_t^{X*} (\omega_{\sigma_t^X(p)}) - \omega_p}{t}$$

Theorem 8.3 (Lie Derivative of Functions).

This is essentially a rephrase of [Definition 3.12](#), but this time we see f as a form.

$$\mathcal{L}_X f = Xf.$$

Proof.

$$\mathcal{L}_X f = \frac{d}{dt} (\sigma_t^{X*} f) \Big|_{t=0} = \frac{d}{dt} (f \circ \sigma_t^X) \Big|_{t=0} = Xf.$$

$\square$

Theorem 8.4 (Various Product Rules).

$$\begin{aligned}\mathcal{L}_X(f\omega) &= (\mathcal{L}_X f)\omega + f\mathcal{L}_X\omega \\ \mathcal{L}_X(\alpha \otimes \beta) &= (\mathcal{L}_X\alpha) \otimes \beta + \alpha \otimes (\mathcal{L}_X\beta).\end{aligned}$$

Proof.

□

Corollary 8.4.1.

$$\mathcal{L}_X(\alpha \wedge \beta) = (\mathcal{L}_X\alpha) \wedge \beta + \alpha \wedge (\mathcal{L}_X\beta).$$

Theorem 8.5 (Lie Derivative of Forms Paired with Vector Fields).

$$\begin{aligned}\mathcal{L}_X(\omega(X_1, \dots, X_k)) \\ = (\mathcal{L}_X\omega)(X_1, \dots, X_k) + \omega(\mathcal{L}_X X_1, \dots, X_k) + \dots + \omega(X_1, \dots, \mathcal{L}_X X_k).\end{aligned}$$

Corollary 8.5.1 (Lie Derivative of Forms as Directional Derivatives and Commutators).

$$\begin{aligned}(\mathcal{L}_X\omega)(X_1, \dots, X_k) \\ = X(\omega(X_1, \dots, X_k)) - \omega([X, X_1], \dots, X_k) - \dots - \omega(X_1, \dots, [X, X_k]).\end{aligned}$$

Theorem 8.6 (Cartan's Magic Formula).

$$\mathcal{L}_X\omega = X \lrcorner (d\omega) + d(X \lrcorner \omega).$$

Corollary 8.6.1 (Lie Derivative Commute with Exterior Derivative).

$$\mathcal{L}_X(d\omega) = d(\mathcal{L}_X\omega).$$

9 Lie Groups and Lie Algebras

9.1 Definitions

Definition 9.1 (Lie Groups).

A Lie group G is a smooth manifold endowed with a group structure.

Definition 9.2 (Translation).

In a Lie group G , we define the left translation by g , denoted $L_g : G \rightarrow G$, and right translation by g , denoted $R_g : G \rightarrow G$, as

$$L_g(h) = gh, \quad R_g(h) = hg.$$

Definition 9.3 (Lie Algebra).

A vector space V together with a map $[\cdot, \cdot] : V \times V \rightarrow V$, is called a Lie algebra if the map is bilinear, antisymmetric and satisfies the Jacobi identity,

$$[u, [v, w]] + [v, [w, u]] + [w, [u, v]] = 0.$$

in which case the map is called the Lie bracket.

Definition 9.4 (Lie Algebra Homomorphism).

Let $(V, [\cdot, \cdot]_V)$ and $(W, [\cdot, \cdot]_W)$ be Lie algebras. A linear map $\psi : V \rightarrow W$ is called Lie algebra homomorphism if

$$[\psi(x), \psi(y)]_W = \psi([x, y]_V), \quad x, y \in V.$$

✍ Remark.

In this regard, the set of smooth vector fields on a manifold $\mathcal{M}$ forms a Lie algebra, which is infinite-dimensional if $\dim \mathcal{M} \geq 1$. It is not that easy to handle.

As it happens, there is a special Lie subalgebra for every Lie group that has the same dimension as the Lie group itself. We usually refer to that specific subalgebra as “the” Lie algebra. ✎

9.2 Invariant Vector Fields

Definition 9.5 (Invariant Vector Fields).

Given a manifold $\mathcal{M}$ and a set of diffeomorphisms Γ from $\mathcal{M}$ to $\mathcal{M}$, we define the set of vector fields which are invariant under Γ as

$$A_\Gamma(\mathcal{M}) = \{X \mid \phi_*X = X, \forall \phi \in \Gamma\}.$$

Theorem 9.1 (Invariant Vector Fields Form Subalgebra).

For any set Γ of diffeomorphisms of $\mathcal{M}$, the set of invariant vector fields under Γ is a Lie subalgebra in the Lie algebra of vector fields endowed with the commutator.

Proof. Suppose Γ consists of a single diffeomorphism ϕ . If X, Y are both ϕ -invariant, for all $a, b \in \mathbb{R}$,

$$\phi_*(aX + bY) = a\phi_*X + b\phi_*Y = aX + bY.$$

Also

$$\phi_*[X, Y] = [\phi_*X, \phi_*Y] = [X, Y].$$

Therefore, for those Γ that contains multiple diffeomorphisms, we can just say

$$A_\Gamma(\mathcal{M}) = \bigcap_{\phi \in \Gamma} A_\phi(\mathcal{M}).$$

□

Definition 9.6 (Left-Invariant Vector Fields).

We define the left-invariant vector fields to be the set

$$L_{g*}X = X, \quad \forall g \in G.$$

9.3 The Associated Lie Algebra

Definition 9.7 (Lie Algebra of a Lie Group).

We define the set of left-invariant vector fields together with the commutator of vector fields on a Lie group G to be the Lie algebra of G , denoted by $\mathfrak{g}$. It is guaranteed by the previous theorem that it is indeed a subalgebra.

Theorem 9.2 (Lie Algebra and Tangent Space).

Let G be a Lie group with identity element e and Lie algebra $\mathfrak{g}$. The evaluation map $\text{ev} : \mathfrak{g} \rightarrow T_e G$ where $\text{ev}(X) = X_e$ is a linear isomorphism.

Proof. Explicitly construct the inverse of the evaluation map, $\iota : T_e G \rightarrow \mathfrak{g}$ by $A \mapsto L_{g*} A := \tilde{A}_g$.

(Injective) If $\tilde{A} = \tilde{B}$, then $A = \tilde{A}_e = \tilde{B}_e = B$.

(Surjective) We require $\forall X \in \mathfrak{g}, \exists A \in T_e G$ s.t. $X_e = A$ which is clearly the case. $\square$

Corollary 9.2.1 (Dimension of Lie Algebra).

Since the Lie algebra $\mathfrak{g}$ of Lie group G is isomorphic to $T_e G$, we have

$$\dim \mathfrak{g} = \dim T_e G = \dim G.$$

Theorem 9.3 (Lie's Third Theorem).

Every finite-dimensional real Lie algebra is isomorphic to the Lie algebra of some connected Lie group.

Remark.

See the remark after [Theorem 9.11](#).

$\mathfrak{M}$

9.4 Induced Homomorphism

Definition 9.8 (Induced Homomorphism of Lie Algebras).

Let G, H be Lie groups, and $\phi : G \rightarrow H$ a homomorphism of Lie groups. Given any $X \in \mathfrak{g}$, we can uniquely define a left-invariant vector field $\tilde{\phi}_* X \in \mathfrak{h}$ by

$$(\tilde{\phi}_* X)_e = \phi_* X_e.$$

The map $\tilde{\phi}_* : \mathfrak{g} \rightarrow \mathfrak{h}$ is called the induced homomorphism of the homomorphism ϕ .

Theorem 9.4 (Induced Homomorphism Is a Homomorphism).

The induced homomorphism $\tilde{\phi}_* : \mathfrak{g} \rightarrow \mathfrak{h}$ is a homomorphism of Lie algebras given a Lie group homomorphism $\phi : G \rightarrow H$.

Proof. We need to show

$$[\tilde{\phi}_*X, \tilde{\phi}_*Y] = \tilde{\phi}_*[X, Y].$$

If $\tilde{\phi}_*X$ is actually ϕ -related to X , then that result is immediate. (By the previous result [Theorem 3.8](#).) Since the defining relation says $(\tilde{\phi}_*X)_e = \phi_*(X_e)$, we can say

$$\begin{aligned} (\tilde{\phi}_*X)_{\phi(g)} &= L_{\phi(g)*}((\tilde{\phi}_*X)_e) \\ &= L_{\phi(g)*}(\phi_*(X_e)) \\ &= (L_{\phi(g)} \circ \phi)_*X_e \end{aligned}$$

Now $\phi(g)\phi(h) = \phi(gh)$, so

$$\begin{aligned} (\tilde{\phi}_*X)_{\phi(g)} &= (\phi \circ L_g)_*X_e \\ &= \phi_*L_{g*}X_e \\ &= \phi_*X_g \end{aligned}$$

□

Corollary 9.4.1 (Subgroup has Subalgebra).

Let $H \subseteq G$ be an immersed or embedded Lie subgroup. Then the Lie algebra $\mathfrak{h} \subseteq \mathfrak{g}$ is a Lie subalgebra.

Proof. The inclusion $i : H \rightarrow G$ is a Lie group homomorphism and an immersion. therefore the induced inclusion $i_* : \mathfrak{h} \rightarrow \mathfrak{g}$ is an injective homomorphism of Lie algebras. □

✍ Remark.

Though at this point I haven't yet grasped the true meaning when others say that "Lie algebra represents infinitesimal actions." But believing this perspective, there is a theorem that connects the infinitesimal, linear Lie algebra homomorphism and the macroscopic, non-linear Lie group homomorphism—the so called Integrability theorem.

As for the problem that whether all Lie subalgebra comes from a Lie subgroup, the proof requires foliations. ✎

Theorem 9.5 (Lie Subalgebra Comes From Subgroups).

Given a Lie group G with associated Lie algebra $\mathfrak{g}$, any subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$ comes from some connected Lie subgroup $H \subseteq G$.

Theorem 9.6 (Integrability for Lie Algebra Homomorphisms).

Let G be a connected and simply connected Lie group, H a Lie group and $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$ a Lie algebra homomorphism. Then there exists a unique Lie group homomorphism $\psi : G \rightarrow H$ s.t. $\psi_* = \phi$.

9.5 Lie Algebra of General Linear Group

Theorem 9.7 (The Shape of GL and $\mathfrak{gl}$).

The Lie group $G = GL(n, \mathbb{R})$ is an open subset of $\mathbb{R}^{n^2}$. Therefore,

$$T_I GL(n, \mathbb{R}) \simeq \mathfrak{gl}(n, \mathbb{R}) \simeq \mathbb{R}^{n^2} = M_{n \times n}(\mathbb{R}).$$

Theorem 9.8 (Left-Invariant Vector Fields in GL).

If $X \in M_{n \times n}(\mathbb{R}) = \mathfrak{gl}(n, \mathbb{R})$, then the associated left-invariant vector field $\tilde{X}$ on $GL(n, \mathbb{R})$ is given by

$$\tilde{X}_A = AX, \quad \forall A \in GL(n, \mathbb{R}).$$

Proof. Since $\mathfrak{gl}(n, \mathbb{R}) \simeq T_I GL(n, \mathbb{R})$, we can also regard X as a vector in $T_I GL(n, \mathbb{R})$. Choose an arbitrary $\sigma^X : I \subseteq \mathbb{R} \rightarrow GL(n, \mathbb{R})$ s.t. $\sigma^X(0) = I$ and $(\sigma^X)'(0) = X$. Then

$$\begin{aligned} \tilde{X}_A &= L_{A*} X \\ &= \left. \frac{d}{dt} (L_A \circ \sigma^X) \right|_{t=0} \\ &= A \left. \frac{d}{dt} \sigma^X \right|_{t=0} \\ &= AX. \end{aligned}$$

□

Theorem 9.9 (Alternative Expression of Lie Bracket).

Let X, Y be vector fields on an open subset $U \subseteq \mathbb{R}^N$, and σ^X, σ^Y are local flows of X and Y , respectively. Then

$$[X, Y]_p = \left. \frac{d}{dt} Y_{\sigma^X(t)} \right|_{t=0} - \left. \frac{d}{dt} X_{\sigma^Y(t)} \right|_{t=0}.$$

Proof. Using the functional product rule [Theorem 3.7](#), we have

$$\begin{aligned} [X, Y] &= [X^\mu \partial_\mu, Y^\nu \partial_\nu] \\ &= X^\mu Y^\nu [\partial_\mu, \partial_\nu] + X^\mu (\partial_\mu Y^\nu) \partial_\nu - Y^\nu (\partial_\nu X^\mu) \partial_\mu \end{aligned}$$

But since we are on Euclidean space with the standard frame, we have $[\partial_\mu, \partial_\nu] = 0$. Also the indices ν and μ is interchangeable, for they are of the same range,

$$\begin{aligned} [X, Y] &= X^\mu (\partial_\mu Y^\nu) \partial_\nu - Y^\nu (\partial_\nu X^\mu) \partial_\mu \\ &= (XY^\nu - YX^\nu) \partial_\nu \\ &= \left(\left. \frac{d}{dt} (Y^\nu \circ \sigma^X) \right|_{t=0} - \left. \frac{d}{dt} (X^\nu \circ \sigma^Y) \right|_{t=0} \right) \partial_\nu \\ &= \left. \frac{d}{dt} Y_{\sigma^X} \right|_{t=0} - \left. \frac{d}{dt} X_{\sigma^Y} \right|_{t=0}. \end{aligned}$$

□

Theorem 9.10 (The Lie Algebra of General Linear Groups).

The Lie algebra of the general linear group $GL(n, \mathbb{R})$ is $\mathfrak{gl}(n, \mathbb{R}) = M_{n \times n}(\mathbb{R})$, and the Lie bracket on $\mathfrak{gl}(n, \mathbb{R})$ is given by the standard commutator of matrices,

$$[X, Y] = XY - YX, \quad \forall X, Y \in \mathfrak{gl}(n, \mathbb{R}).$$

Proof. Since $GL(n, \mathbb{R})$ is an open subset of Euclidean space, the requirements of [Theorem 9.9](#) are satisfied. Then

$$\begin{aligned} [X, Y]_I &= \left. \frac{d}{dt} Y_{\sigma^X(t)} \right|_{t=0} - \left. \frac{d}{dt} X_{\sigma^Y(t)} \right|_{t=0} \\ &= \left. \frac{d}{dt} \sigma^X(t) Y \right|_{t=0} - \left. \frac{d}{dt} \sigma^Y(t) X \right|_{t=0} \\ &= XY - YX. \end{aligned}$$

□

Theorem 9.11 (Ado's Theorem).

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over $\mathbb{R}$ or $\mathbb{C}$. Then there exists an injective Lie algebra homomorphism of $\mathfrak{g}$ into $\mathfrak{gl}(n, \mathbb{R})$ or $\mathfrak{gl}(n, \mathbb{C})$ for some n .

✍ Remark.

Ado's Theorem **Theorem 9.11** says that every $\mathfrak{g}$ over $\mathbb{R}$ can be injectively homomorphed into $\mathfrak{gl}(n, \mathbb{R})$, which means the range of that homomorphism must be a subalgebra of $\mathfrak{gl}(n, \mathbb{R})$.

Now since every $\mathfrak{h} \subseteq \mathfrak{g}$ comes from a connected Lie subgroup $H \subseteq G$, these two results imply Lie's third theorem, which said any finite-dimensional Lie algebra over $\mathbb{R}$ is isomorphic to the Lie algebra of some connected Lie group—that “some connected Lie group” can be a subgroup of the general linear group. $\square$

9.6 The Exponential Map

9.6.1 Properties of Left-Invariant Integral Curves

In this section, we denote G a Lie group, $\mathfrak{g}$ its Lie algebra, $X \in \mathfrak{g}$ a left-invariant vector field, and its maximal integral curve passing through the identity $\phi_X : I = (t_{\min}, t_{\max}) \subseteq \mathbb{R} \rightarrow G$. By definition, we have $\phi_X(0) = e$, and $X_{\phi_X(t)} = [\phi_X(t)]$.

Theorem 9.12 (Left-Invariant Integral Curve is Lie Homomorphism).

$$\phi_X(s+t) = \phi_X(s)\phi_X(t), \quad \forall s, t \in \mathbb{R}.$$

Proof. First fix $s \in \mathbb{R}$ and therefore $g = \phi_X(s) \in G$ is fixed also. Consider $\eta(t) : I \rightarrow G$ by $\eta(t) = g\phi_X(t)$ and $\tilde{\eta} : (t_{\min} - s, t_{\max} - s) \rightarrow G$ by $\tilde{\eta}(t) = \phi_X(s+t)$. Then they are both integral curves of X with $\eta(0) = \tilde{\eta}(0) = g$. So on the intersection of their domains, they are equal by the uniqueness of integral curves. $\square$

Theorem 9.13 (Maximality of Left-Invariant Integral Curve).

ϕ_X is defined on all of $\mathbb{R}$. That is, $t_{\min} = -\infty$ and $t_{\max} = \infty$.

Proof. Suppose $t_{\max} < \infty$ and set $\alpha = \min\{t_{\max}, |t_{\min}|\} < \infty$. Consider $\gamma : (-\frac{\alpha}{2}, \frac{3\alpha}{2}) \rightarrow G$ by $\gamma(t) = \phi_X(\frac{\alpha}{2})\phi_X(t - \frac{\alpha}{2})$. Then $\gamma(0) = e$, and it is an integral curve of X by its homomorphism property.

But then $\frac{3\alpha}{2} > t_{max}$, contradicting the fact that t_{max} is the maximal choice of domain. So $t_{max} = \infty$. $\square$

Theorem 9.14 (Exponential Property of Left-Invariant Integral Curve).

$$\phi_{sX}(t) = \phi_X(st), \quad \forall s, t \in \mathbb{R}.$$

Proof. Consider the curve $\delta : \mathbb{R} \rightarrow G$ by $\delta(t) = \phi_X(st)$. It is easy to show $\delta(0) = e$, and δ is an integral curve of sX . So the result follows by uniqueness of integral curves. $\square$

9.6.2 The Exponential Map

Definition 9.9 (The Exponential Map).

Let $\phi_X : \mathbb{R} \rightarrow G$ be the integral curve through $e \in G$ for $X \in \mathfrak{g}$. Then we define the exponential map $\exp : \mathfrak{g} \rightarrow G$ by

$$\exp(X) = \phi_X(1).$$

Theorem 9.15 (Properties of the Exponential Map).

$$\begin{aligned} \exp(0) &= e \\ \exp((s+t)X) &= \exp(sX) \exp(tX) \\ \exp(-X) &= (\exp X)^{-1}. \end{aligned}$$

9.6.3 Flow and Exponential Map

Theorem 9.16 (The Flow and the Exponential Map).

Let G be a Lie group and $X \in \mathfrak{g}$ a left-invariant vector field. Then its flow $\phi_t(p)$ through a point $p \in G$ is defined for all $t \in \mathbb{R}$, and is given by

$$\phi_t(p) = p \exp tX = L_p(\exp tX).$$

Proof. We let $\phi_t(p) = p \exp tX$ and show it is indeed the flow of X . Firstly, $\phi_0(p) =$

$p \exp(0) = pe = p$. Then, consider the curve $\phi_t(p)$ at $t = s$,

$$\begin{aligned} [\phi_t(p)]_{t=s} &= [p \exp tX]_{t=s} \\ &= L_{p*}[\exp tX]_{t=s} \\ &= L_{p*}X_{\exp sX} \\ &= X_{p \exp sX} \\ &= X_{\phi_s(p)}. \end{aligned}$$

By the uniqueness of integral curves, the claim was proved. $\square$

Theorem 9.17 (Exponential Map and Induced Homomorphisms).

Let $\psi : G \rightarrow H$ be a homomorphism between Lie groups and $\psi_* : \mathfrak{g} \rightarrow \mathfrak{h}$ be the induced homomorphism on Lie algebras. Then

$$\psi(\exp X) = \exp(\psi_* X), \quad X \in \mathfrak{g}.$$

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{\psi_*} & \mathfrak{h} \\ \exp \downarrow & & \downarrow \exp \\ G & \xrightarrow{\psi} & H \end{array}$$

Figure 1: Commutative diagram of Exponential map and homomorphisms between Lie Algebras.

Proof. Consider the curve $\sigma(t) = \psi(\exp tX)$. We see $\sigma(0) = \psi(e) = e$. Now, consider

$$\begin{aligned} [\sigma]_{t=s} &= [\psi \circ \exp tX]_{t=s} \\ &= \psi_*[\exp tX]_{t=s} \\ &= \psi_*X_{\exp sX} \\ &= \psi_*L_{\exp sX*}X_e \\ &= (\psi \circ L_{\exp sX})_*X_e \\ &= (L_{\psi(\exp sX)} \circ \psi)_*X_e \\ &= L_{\sigma(s)*}(\psi_*X) \\ &= (\psi_*X)_{\sigma(s)} \end{aligned}$$

because both X and ψ_*X are left-invariant, and ψ is a Lie group homomorphism. We see from the result that $\sigma(t)$ is an integral curve of ψ_*X through the point e . By the

uniqueness of integral curve,

$$\sigma(1) = \exp(\psi_* X) = \psi(\exp X).$$

□

9.6.4 The Matrix Exponential Map

Definition 9.10 (The Matrix Exponential).

Let $A \in M_{n \times n}$ be a square matrix over either reals or complex numbers. Then we define

$$e^A = \sum_{k=0}^{\infty} \frac{1}{k!} A^k.$$

This series always converge.

Theorem 9.18 (Exponential of Sum for Commuting Matrices).

If $A, B \in M_{n \times n}$ commutes, i.e., $AB = BA$, then

$$e^{A+B} = e^A e^B.$$

Theorem 9.19 (Properties of Matrix Exponential).

For all $A \in M_{n \times n}$, the map $\phi_A : \mathbb{R} \rightarrow GL(n, K)$ by $\phi_A(t) = e^{tA}$ is smooth and satisfies

$$\phi_A(0) = I, \quad \left. \frac{d}{dt} \phi_A(t) \right|_{t=s} = \phi_A(s) A \quad \forall s \in \mathbb{R}.$$

Proof. We have

$$\begin{aligned} \left. \frac{d}{dt} \phi_A(t) \right|_{t=s} &= \lim_{h \rightarrow 0} \frac{\phi_A(s+h) - \phi_A(s)}{h} \\ &= \lim_{h \rightarrow 0} \frac{e^{sA}(e^{hA} - I)}{h} \\ &= e^{sA} \lim_{h \rightarrow 0} \frac{e^{hA} - I}{h}. \end{aligned}$$

Now we know

$$e^{hA} = I + hA + \frac{1}{2!} h^2 A^2 + \dots$$

and this yields the result.

□

Theorem 9.20 (Exponential Map of Linear Group is Matrix Exponential).

Let $A \in \mathfrak{gl}(n, \mathbb{R})$ or $\mathfrak{gl}(n, \mathbb{C})$. Then

$$\exp(A) = e^A.$$

Theorem 9.21 (Determinant of Matrix Exponential).

Let $X \in M_{n \times n}$. Then

$$\det(e^X) = e^{\operatorname{tr}(X)}.$$

Proof. Consider a curve $\sigma : I \subseteq \mathbb{R} \rightarrow GL(n, \mathbb{R})$ by $\sigma(t) = I + tX$. We see $\sigma(0) = I$, and $[\sigma]_{t=0} = X$ (Using the canonical identification). Now,

$$\begin{aligned} \det_* \sigma &= \left[\det(\sigma(t)) \right]_{t=0} \\ &= [\det(I + tX)]_{t=0}. \end{aligned}$$

Using the canonical identification,

$$\begin{aligned} [\det(I + tX)]_{t=0} &= \left. \frac{d}{dt} \det(I + tX) \right|_{t=0} \\ &= \left. \frac{d}{dt} (1 + \operatorname{tr}(X)t + O(t^2)) \right|_{t=0} \\ &= \operatorname{tr}(X). \end{aligned}$$

Now using the commuting diagram of **Theorem 9.17**, we see

$$\det(e^X) = e^{\det_* X} = e^{\operatorname{tr} X}.$$

□

Theorem 9.22 (Lie Algebras of Classical Groups).

The Lie algebras of the special linear groups are

$$\mathfrak{sl}(n, \mathbb{R}) = \{M \in M_{n \times n}(\mathbb{R}) \mid \operatorname{tr}(M) = 0\}$$

$$\mathfrak{sl}(n, \mathbb{C}) = \{M \in M_{n \times n}(\mathbb{C}) \mid \operatorname{tr}(M) = 0\}.$$

The Lie algebras of the orthogonal and unitary groups are

$$\mathfrak{o}(n, \mathbb{R}) = \{M \in M_{n \times n}(\mathbb{R}) \mid M + M^\top = 0\}$$

$$\mathfrak{u}(n) = \{M \in M_{n \times n}(\mathbb{C}) \mid M + M^\dagger = 0\}.$$

The Lie algebras of the special orthogonal and special unitary groups are

$$\mathfrak{so}(n) = \mathfrak{o}(n)$$

$$\mathfrak{su}(n) = \{M \in M_{n \times n}(\mathbb{C}) \mid M + M^\dagger = 0, \operatorname{tr} M = 0\}.$$